

Quasi-maximum likelihood estimation for scalable ARMA models

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June 4, 2026

Abstract

The recently proposed scalable ARMA model preserves the parsimony of traditional VARMA models while achieving greater computational tractability. However, existing studies are limited to regularized least squares estimation (LSE) for high-dimensional settings, which is not only statistically less efficient but also requires the sub-Gaussian assumption for its theoretical guarantees. Moreover, it still lacks inference tool for real applications. To fill this gap, we develop a quasi-maximum likelihood estimation (QMLE) framework for scalable ARMA models. Its asymptotic normality is established under a finite fourth order moment condition, and we formally prove its asymptotic efficiency gain over LSE. We also introduce an efficient block coordinate descent algorithm for computation and a consistent Bayesian information criterion for model selection. Simulation studies validate the finite-sample performance of our methodology, and an empirical application to six macroeconomic indicators demonstrates its practical utility.

Keywords: Model selection, quasi-maximum likelihood estimation, scalable ARMA, vector ARMA.

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1 Introduction

Multivariate time series analysis constitutes a fundamental methodology in fields such as economics, finance, and environmental studies ([Lütkepohl, 2005](#)). Economists rely on the tracking of key variables, such as industrial production, civilian unemployment rate, consumer price index, real wages, real personal consumption expenditures, and federal funds rate, to evaluate economic conditions and guide policy or investment decisions. Critically, these economic variables are usually cross-correlated, and understanding their interdependence will help reveal underlying linkage relationships and co-movements within the economic system. In response to this need, several multivariate models have been introduced to capture these dynamic interrelations and enhance forecast accuracy. Notable among these are the vector autoregressive (VAR), vector moving average (VMA), and vector autoregressive moving average (VARMA) ([Sims, 1980](#); [Tiao and Box, 1981](#); [Tiao and Tsay, 1989](#); [Reinsel, 2003](#)). The VAR model is the most widely adopted owing to its theoretical simplicity, computational ease, and clear interpretation ([Tsay, 2014](#)). Despite its popularity, a practical limitation of the VAR model is that it often necessitates a high lag order to achieve a satisfactory fit. This results in a large number of coefficient matrices and thus a lack of parsimony ([Chan et al., 2016](#)).

Compared to VAR models, the VARMA model has a more parsimonious structure. Meanwhile, the invertible VARMA model is equivalent to a $\text{VAR}(\infty)$ form, which enables it to capture temporal dependence in a more flexible and comprehensive manner than finite-order VAR models. This flexibility contributes to superior forecasting performance of VARMA models over VAR models, as supported by much empirical evidence ([Poskitt, 2003](#); [Athanasopoulos and Vahid, 2008](#); [Athanasopoulos et al., 2012](#); [Kascha, 2012](#); [Simionescu, 2013](#); [Dufour and Pelletier, 2022](#)). However, the practical application of VARMA models faces challenges due to issues of non-identifiability and computational complexity. Many studies have attempted to address these issues. For example, various sophisticated identification constraints have been proposed, including the final equations form ([Zellner et al., 1974](#); [Wallis, 1977](#)), the Echelon form ([Hannan and Kavalieris, 1984](#); [Lütkepohl and Poskitt, 1996](#)),

and scalar component models (Tiao and Tsay, 1989). To mitigate estimation difficulties, an iterative ordinary least squares method was introduced in Dias and Kapetanios (2018). Moreover, Chan et al. (2016) developed a Bayesian approach to ensure identification and parsimony in VARMA modeling, and Wilms et al. (2023) established a convex optimization-based framework for identification and estimation. Nevertheless, these approaches primarily address the identification problem by imposing complex identification conditions or tackle the computational intractability by optimization algorithms, without fundamentally addressing the inherent limitations rooted in the model’s construction.

Recent work by Zheng (2025) and Huang et al. (2025) proposed a scalable ARMA model formulated in a VAR(∞) form. This model retains the parsimony and rich dynamic patterns (e.g., exponential and damped sinusoidal decays) of traditional VARMA models, while successfully circumventing their inherent issues of non-identifiability and computational intractability. Its VAR(∞) representation also allows for direct inference of multivariate Granger causality through its coefficient matrices. Nonetheless, two critical gaps remain. First, existing work is confined to regularized least squares estimation (LSE) for high-dimensional data, supported by non-asymptotic error bounds. These bounds, however, do not facilitate standard statistical inference such as constructing confidence intervals or significance tests. Consequently, valid inferential tools for scalable ARMA models remain unknown. Moreover, given the prominent role of factor models in dimension reduction, integrating inference for scalable ARMA with factor structures could offer powerful tools for high-dimensional analysis. Second, LSE itself is statistically less efficient, as it ignores the contemporaneous correlations and heterogeneity among innovations. Moreover, non-asymptotic theories require the restrictive sub-Gaussian assumption. This limits its applicability to heavier-tailed economic and financial time series. To address these gaps, this paper develops a more efficient and robust inference framework for scalable ARMA models under a finite fourth order moment condition. The proposed tools enable a deeper understanding of the interdependence and dynamic patterns in multiple series, as illustrated in the application (Section 6).

Our main contributions can be summarized as follows: (i) we develop the quasi-maximum

likelihood estimation (QMLE) for the scalable ARMA model, establishing its asymptotic properties under relaxed moment conditions. The asymptotic efficiency of the QMLE over the least squares estimation (LSE) is also rigorously confirmed; (ii) we propose a block coordinate descent (BCD) algorithm for the efficient computation of the QMLEs; (iii) we introduce an asymptotically consistent Bayesian information criterion (BIC) for model order selection. Moreover, simulation studies validate these theoretical results. Furthermore, the practical value of our proposed framework is highlighted by an empirical study on exploring the Granger-causal relationship among six macroeconomic variables, as well as the forecasting comparison with VAR and VARMA models.

The remainder of this paper is organized as follows. Section 2 introduces the scalable ARMA model. Section 3 proposes the QMLE method and BCD algorithm for estimation, as well as the BIC method for order selection. Section 4 establishes the asymptotic properties for the proposed QMLE and BIC method, and the asymptotic efficiency of the QMLE and LSE is also compared. Simulation studies and an empirical example are provided in Sections 5 and 6, respectively. The conclusion and discussion appear in Section 7. Technical proofs of all theorems and propositions are provided in the Appendix. Throughout the paper, we denote vectors by small boldface letters and matrices by capital letters. Denote $\|\mathbf{x}\|_p$ as the ℓ_p norm of a vector $\mathbf{x}$, that is $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$. For a matrix A , $\|A\|_{\text{op}}$, $\|A\|$, A' and $\text{vec}(A)$ are the operator norm (i.e., $\|A\|_{\text{op}} = \sup_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|$), matrix norm satisfying sub-multiplicative property, transpose and vectorization of A , respectively. For a square matrix B , $|B|$, $\text{tr}(B)$ and $\text{vech}(B)$ are the determinant, trace, and the lower triangular-stacking vector of B , respectively. And $B \succeq 0$ means that B is semi-positive definite. For square matrices B_j 's, let $\text{diag}\{B_1, \dots, B_m\}$ be the block diagonal matrix whose main diagonal consists of $B_1, \dots, B_m$. Moreover, $\mathbf{1}_N$ is an $N \times 1$ vector of ones, $0_{N \times T}$ is an $N \times T$ matrix of zeroes, and I_N is an $N \times N$ identity matrix. Additionally, $O(1)$ denotes a bounded series of nonstochastic variables, $\rightarrow_p$ denotes the convergence in probability, and $\rightarrow_d$ denotes the convergence in distribution. The data that support the findings of this study in Section 6 are openly available in FRED-MD repository at stlouisfed.org/research/economists/mccracken/fred-databases, and computer programs for the analysis are available at github.com/

2 Scalable ARMA models

Following [Zheng \(2025\)](#) and [Huang et al. \(2025\)](#), the scalable ARMA model for an N -dimensional process $\mathbf{y}_t$ is given by the parametric VAR(∞) form:

$$\mathbf{y}_t = \sum_{h=1}^{\infty} A_h(\boldsymbol{\omega}, \mathbf{g}) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t = \sum_{k=1}^d G_k \sum_{h=1}^{\infty} \ell_{hk}(\boldsymbol{\omega}) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t, \quad (2.1)$$

where $\{\boldsymbol{\varepsilon}_t\}$ is a sequence of independent and identically distributed (*i.i.d.*) random vectors with zero mean and covariance matrix $\Sigma = (\sigma_{ij})_{1 \leq i, j \leq N}$, $d = p + r + 2s$, $\boldsymbol{\omega} = (\boldsymbol{\lambda}', \boldsymbol{\eta}')$, $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_r)'$ with $|\lambda_i| \in (0, 1)$ for $1 \leq i \leq r$, $\boldsymbol{\eta} = (\boldsymbol{\eta}'_1, \dots, \boldsymbol{\eta}'_s)'$ with $\boldsymbol{\eta}_j = (\gamma_j, \varphi_j) \in (0, 1) \times (0, \pi)$ for $1 \leq j \leq s$, $\mathbf{g} = \text{vec}(G_1, \dots, G_d)$ with $G_k \in \mathbb{R}^{N \times N}$ for $1 \leq k \leq d$, and the entries $\ell_{hk}(\boldsymbol{\omega})$ for $h \geq 1$ and $1 \leq k \leq d$ are defined as follows:

$$\ell_{hk}(\boldsymbol{\omega}) = \begin{cases} 1, & \text{if } 1 \leq h = k \leq p, \\ \lambda_i^{h-p}, & \text{if } h > p, \ k = p + i \text{ with } 1 \leq i \leq r, \\ \gamma_j^{h-p} \cos[(h-p)\varphi_j], & \text{if } h > p, \ k = p + r + 2j - 1 \text{ with } 1 \leq j \leq s, \\ \gamma_j^{h-p} \sin[(h-p)\varphi_j], & \text{if } h > p, \ k = p + r + 2j \text{ with } 1 \leq j \leq s, \\ 0, & \text{otherwise.} \end{cases}$$

Particularly, (p, r, s) is the order of model (2.1). To ensure the uniqueness of model (2.1), we assume that $\lambda_1 > \dots > \lambda_r$ and $\gamma_1 > \dots > \gamma_s$. Model (2.1) provides a time series model comparable to the VARMA model, while successfully circumventing the latter's inherent issues of non-identifiability and computational intractability.

For $1 \leq h \leq p$, $\ell_{hk}(\boldsymbol{\omega}) = 1$ if $k = h$ and 0 otherwise, which implies that $A_h = G_h$. For $h > p$, the coefficient $\ell_{hk}(\boldsymbol{\omega})$ follows two distinct decay patterns as h increases: exponential decay and exponentially damped cosine (or sine) waves. In this case, A_h can be decomposed as $A_h = G_h^I + G_h^{II}$, where $G_h^I = \sum_{i=1}^r \lambda_i^{h-p} G_{p+i}$ and $G_h^{II} = \sum_{j=1}^s \gamma_j^{h-p} \{\cos[(h-p)\varphi_j] G_{p+r+2j-1} +$

$\sin[(h-p)\varphi_j]G_{p+r+2j}\}$ capture these two decay patterns, respectively. Consequently, model (2.1) can be rewritten as

$$\mathbf{y}_t = \sum_{h=1}^p G_h \mathbf{y}_{t-h} + \sum_{h=p+1}^{\infty} (G_h^I + G_h^{II}) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t. \quad (2.2)$$

Model (2.2) explicitly separates the short-run and long-run autoregressive effects. Specifically, $\sum_{h=1}^p G_h \mathbf{y}_{t-h}$ captures the short-run effects, analogous to a standard VAR(p) model. In contrast, $\sum_{h=p+1}^{\infty} (G_h^I + G_h^{II}) \mathbf{y}_{t-h}$ captures the long-run effects. The term G_h^I embodies exponential decay via λ_i^{h-p} , while the term G_h^{II} generates exponentially damped oscillatory patterns via $\gamma_j^{h-p} \cos[(h-p)\varphi_j]$ and $\gamma_j^{h-p} \sin[(h-p)\varphi_j]$. Both ensure that long-run contributions vanish fast as the time lag h increases. In addition, we refer to λ_i 's and γ_j 's the decay rate coefficients, φ_j 's the periodic coefficients. In the special case where $r = s = 0$, we have $G_h^I = G_h^{II} = 0_{N \times N}$ for all $h > p$, and the model (2.2) will reduce to a VAR(p) process, retaining only the short-run component.

On the other hand, model (2.1) can also be rewritten as follows

$$\mathbf{y}_t = \sum_{l=1}^p G_l \mathbf{y}_{t-l} + \sum_{i=1}^r G_{p+i} \mathbf{f}^I(\mathbf{x}_t; \lambda_i) + \sum_{j=1}^s \sum_{m=1}^2 G_{p+r+2(j-1)+m} \mathbf{f}^{II,m}(\mathbf{x}_t; \boldsymbol{\eta}_j) + \boldsymbol{\varepsilon}_t, \quad (2.3)$$

where $\mathbf{x}_t = (\mathbf{y}'_{t-1}, \mathbf{y}'_{t-2}, \dots)'$, $\mathbf{f}^I(\mathbf{x}_t; \lambda_i) = \sum_{h=p+1}^{\infty} \lambda_i^{h-p} \mathbf{y}_{t-h}$ for $1 \leq i \leq r$, $\mathbf{f}^{II,1}(\mathbf{x}_t; \boldsymbol{\eta}_j) = \sum_{h=p+1}^{\infty} \gamma_j^{h-p} \cos[(h-p)\varphi_j] \mathbf{y}_{t-h}$ and $\mathbf{f}^{II,2}(\mathbf{x}_t; \boldsymbol{\eta}_j) = \sum_{h=p+1}^{\infty} \gamma_j^{h-p} \sin[(h-p)\varphi_j] \mathbf{y}_{t-h}$ for $1 \leq j \leq s$. It can be seen that the orders p and (r, s) are analogous to the AR and MA orders of the VARMA model (Zheng, 2025). Particularly, model (2.3) will reduce to a VAR(p) model when $r = s = 0$, and it can be treated as a VMA model when $p = 0$. Consequently, we refer to $G_1, \dots, G_p$ the AR coefficient matrices, and $G_{p+1}, \dots, G_d$ the MA coefficient matrices of model (2.1) with the order (p, r, s) .

Additionally, we can use model (2.1) to infer the Granger causal (GC) relations between any pair of component series in $\mathbf{y}_t$ (Zheng, 2025). Since G_k captures cross-sectional information associated with a particular sequence $\{\ell_{hk}(\boldsymbol{\omega})\}_{h=1}^{\infty}$, it enables us to identify the decay pattern of GC relations. Denote $g_{i,j,k}$ as the (i, j) th entry of G_k , and $y_{j,t}$ as the j th entry of

$\mathbf{y}_t$. Consider the influence of $\{y_{j,t}\}$ on $\{y_{1,t}\}$ for illustration. The equation for $y_{1,t}$ in model (2.3) includes two conditional mean terms: the first term involves the sum of $g_{1,j,k}y_{j,t-k}$ for $1 \leq k \leq p$, while the second term involves a weighted mixture of r distinct exponential decay rates and s distinct pairs of damped cosine and sine waves to capture the influence beyond lag p . Hence, if the GC relation exists, the lagged influence of $\{y_{j,t}\}$ on $\{y_{1,t}\}$ can be classified into three scenarios: (1) *short-term only*, $g_{1,j,k} \neq 0$ for some $1 \leq k \leq p$, while $g_{1,j,p+1} = \dots = g_{1,j,d} = 0$; (2) *long-term only*, $g_{1,j,1} = \dots = g_{1,j,p} = 0$, while $g_{1,j,k} \neq 0$ for some $p+1 \leq k \leq d$; and (3) *both short-term and long-term influences*, $g_{1,j,k} \neq 0$ for some $1 \leq k \leq p$ and some $p+1 \leq k \leq d$.

This paper focuses on the case of fixed N , thus requiring no sparsity or low-rank conditions on G_k for $1 \leq k \leq d$. This differs from the high-dimensional frameworks studied by Zheng (2025) and Huang et al. (2025). Below, we propose an estimation method for model (2.1) and an information criterion to select its order (p, r, s) .

3 Methodology

3.1 Quasi-maximum likelihood estimation

Suppose that the time series $\{\mathbf{y}_t\}_{t=1}^T$ is generated by model (2.1) with order (p, r, s) . Recall that $\boldsymbol{\omega} = (\boldsymbol{\lambda}', \boldsymbol{\eta}')'$, $\mathbf{g} = \text{vec}(G_1, \dots, G_d)$, and Σ is the covariance matrix of $\{\boldsymbol{\varepsilon}_t\}$. Denote $\boldsymbol{\alpha} = (\boldsymbol{\omega}', \mathbf{g}')'$ and $\boldsymbol{\sigma} = \text{vech}(\Sigma)$. Let $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \boldsymbol{\sigma}')'$ be the parameter vector of model (2.1) and Θ be the parameter space, where Θ is a compact subspace of $\{(-1, 0) \cup (0, 1)\}^r \times \{(0, 1) \times (0, \pi)\}^s \times \mathbb{R}^{N^2d} \times \mathbb{R}^{N(N+1)/2}$. Conditional on $\{\mathbf{y}_1, \dots, \mathbf{y}_T\}$, the negative Gaussian quasi-log-likelihood function (ignoring a constant) is

$$\mathcal{L}(\boldsymbol{\theta}) = T^{-1} \sum_{t=1}^T l_t(\boldsymbol{\theta}) \quad \text{and} \quad l_t(\boldsymbol{\theta}) = \frac{1}{2} \ln |\Sigma(\boldsymbol{\sigma})| + \frac{1}{2} \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}),$$

where $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\omega}, \mathbf{g}) \mathbf{y}_{t-h}$ is the error function, and $\Sigma(\boldsymbol{\sigma})$ denotes the reconstruction of the symmetric matrix Σ from its half-vectorized form $\boldsymbol{\sigma}$. Note that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$, $l_t(\boldsymbol{\theta})$ and $\mathcal{L}(\boldsymbol{\theta})$ depend on observations in the infinite past. We simply set $\mathbf{y}_s = \mathbf{0}_{N \times 1}$ for $s \leq 0$ as

the initial values for $\{\mathbf{y}_s, s \leq 0\}$ in this paper, and denote the resulting functions of $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$, $l_t(\boldsymbol{\theta})$ and $\mathcal{L}(\boldsymbol{\theta})$ as $\tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})$, $\tilde{l}_t(\boldsymbol{\theta})$ and $\tilde{\mathcal{L}}(\boldsymbol{\theta})$, respectively. Then the Gaussian quasi-maximum likelihood estimator (QMLE) can be defined as

$$\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\alpha}}', \hat{\boldsymbol{\sigma}}')' = \arg \min_{\boldsymbol{\theta} \in \Theta} \tilde{\mathcal{L}}(\boldsymbol{\theta}). \quad (3.1)$$

Because we do not assume that $\boldsymbol{\varepsilon}_t$ is Gaussian, $\hat{\boldsymbol{\theta}}$ is called the QMLE. Our theoretical analysis will prove that the effect of the initial values on the estimation is asymptotically negligible. Moreover, the consistency and asymptotic normality of the QMLE $\hat{\boldsymbol{\theta}}$ will be established under regularity conditions in Section 4.1.

3.2 Block coordinate descent algorithm

The objective function $\tilde{\mathcal{L}}(\boldsymbol{\theta})$ in (3.1) is non-convex and highly nonlinear with respect to the argument $\boldsymbol{\theta} = (\boldsymbol{\omega}', \mathbf{g}', \boldsymbol{\sigma}')'$, and thus the QMLE $\hat{\boldsymbol{\theta}}$ has no explicit form and needs to be obtained by optimization algorithm. While $\tilde{\mathcal{L}}(\boldsymbol{\theta})$ is not jointly convex, the parameterization in model (2.3) allows a convenient conditional block structure: (i) for fixed $(\mathbf{g}, \boldsymbol{\sigma})$, it is smooth in $\boldsymbol{\omega}$; (ii) for fixed $(\boldsymbol{\omega}, \boldsymbol{\sigma})$, it is quadratic in $\mathbf{g}$; and (iii) for fixed $(\boldsymbol{\omega}, \mathbf{g})$, it has a closed-form solution for $\boldsymbol{\sigma}$; see details in (3.2)–(3.5). This motivates us to employ a BCD algorithm to solve the QMLE $\hat{\boldsymbol{\theta}}$.

Recall that $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \boldsymbol{\sigma}')' = (\boldsymbol{\omega}', \mathbf{g}', \boldsymbol{\sigma}')'$, and rewrite $\tilde{\mathcal{L}}(\boldsymbol{\theta}) = \tilde{\mathcal{L}}(\boldsymbol{\omega}, \mathbf{g}, \boldsymbol{\sigma})$. The BCD algorithm starts from an initial point $\boldsymbol{\theta}^{(0)} = (\boldsymbol{\alpha}^{(0)'}, \boldsymbol{\sigma}^{(0)'})' = (\boldsymbol{\omega}^{(0)'}, \mathbf{g}^{(0)'}, \boldsymbol{\sigma}^{(0)'})'$, and generates a sequence of vectors $\{\boldsymbol{\theta}^{(k)}\}_{k=0}^\infty$. Denote $\tilde{\mathbf{x}}_t = (\mathbf{y}'_{t-1}, \dots, \mathbf{y}'_1, \mathbf{0})'$. Define the $N \times N^2 d$ matrix $Z_t^{(k+1)} = \{\mathbf{y}'_{t-1}, \dots, \mathbf{y}'_{t-p}, \mathbf{f}^I(\tilde{\mathbf{x}}_t; \lambda_1^{(k+1)})', \dots, \mathbf{f}^I(\tilde{\mathbf{x}}_t; \lambda_r^{(k+1)})', \mathbf{f}^{II,1}(\tilde{\mathbf{x}}_t; \boldsymbol{\eta}_1^{(k+1)})', \dots, \mathbf{f}^{II,1}(\tilde{\mathbf{x}}_t; \boldsymbol{\eta}_s^{(k+1)})', \mathbf{f}^{II,2}(\tilde{\mathbf{x}}_t; \boldsymbol{\eta}_1^{(k+1)})', \dots, \mathbf{f}^{II,2}(\tilde{\mathbf{x}}_t; \boldsymbol{\eta}_s^{(k+1)})'\} \otimes I_N$. Then given $\boldsymbol{\omega}^{(k+1)}$ and initial values $\mathbf{y}_s = \mathbf{0}_{N \times 1}$ for $s \leq 0$, we can rewrite model (2.3) as

$$\mathbf{y}_t = Z_t^{(k+1)} \mathbf{g} + \boldsymbol{\varepsilon}_t.$$

For each outer iteration from $\boldsymbol{\theta}^{(k)}$ to $\boldsymbol{\theta}^{(k+1)}$, we update $\boldsymbol{\omega}, \mathbf{g}$ and $\boldsymbol{\sigma}$ sequentially:

- (i) **Update ω .** Given $\lambda_1^{(k+1)}, \dots, \lambda_{i-1}^{(k+1)}, \lambda_{i+1}^{(k)}, \dots, \lambda_r^{(k)}, \boldsymbol{\eta}^{(k)}, \mathbf{g}^{(k)}$ and $\boldsymbol{\sigma}^{(k)}$, we update $\lambda_i^{(k+1)}$ by

$$\lambda_i^{(k+1)} = \arg \min_{\lambda_i} \tilde{\mathcal{L}}(\lambda_1^{(k+1)}, \dots, \lambda_{i-1}^{(k+1)}, \lambda_i, \lambda_{i+1}^{(k)}, \dots, \lambda_r^{(k)}, \boldsymbol{\eta}^{(k)}, \mathbf{g}^{(k)}, \boldsymbol{\sigma}^{(k)}), \quad (3.2)$$

and given $\boldsymbol{\lambda}^{(k+1)}, \boldsymbol{\eta}_1^{(k+1)}, \dots, \boldsymbol{\eta}_{j-1}^{(k+1)}, \boldsymbol{\eta}_{j+1}^{(k)}, \dots, \boldsymbol{\eta}_s^{(k)}, \boldsymbol{\sigma}^{(k)}$, we update $\boldsymbol{\eta}_j^{(k+1)}$ by

$$\boldsymbol{\eta}_j^{(k+1)} = \arg \min_{\boldsymbol{\eta}_j} \tilde{\mathcal{L}}(\boldsymbol{\lambda}^{(k+1)}, \boldsymbol{\eta}_1^{(k+1)}, \dots, \boldsymbol{\eta}_{j-1}^{(k+1)}, \boldsymbol{\eta}_j, \boldsymbol{\eta}_{j+1}^{(k)}, \dots, \boldsymbol{\eta}_s^{(k)}, \mathbf{g}^{(k)}, \boldsymbol{\sigma}^{(k)}), \quad (3.3)$$

where (3.2) and (3.3) are only one- or two-dimensional, and thus can be solved using the NR algorithm.

- (ii) **Update $\mathbf{g}$.** Given $\boldsymbol{\omega}^{(k+1)}$ and $\boldsymbol{\sigma}^{(k)}$, $\mathbf{g}$ can be updated by the following closed-form:

$$\mathbf{g}^{(k+1)} = \left(\sum_{t=1}^T Z_t^{(k+1)'} (\boldsymbol{\Sigma}^{(k)})^{-1} Z_t^{(k+1)} \right)^{-1} \left(\sum_{t=1}^T Z_t^{(k+1)'} (\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{y}_t \right), \quad (3.4)$$

where $\boldsymbol{\sigma}^{(k)} = \text{vech}(\boldsymbol{\Sigma}^{(k)})$.

- (iii) **Update $\boldsymbol{\sigma}$.** Given $\boldsymbol{\omega}^{(k+1)}$ and $\mathbf{g}^{(k+1)}$, $\boldsymbol{\sigma}^{(k+1)}$ is updated by

$$\boldsymbol{\sigma}^{(k+1)} = \text{vech} \left(\frac{1}{T} \sum_{t=1}^T \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}^{(k+1)}) \tilde{\boldsymbol{\varepsilon}}_t'(\boldsymbol{\alpha}^{(k+1)}) \right). \quad (3.5)$$

Finally, we have $\boldsymbol{\theta}^{(k+1)} = (\boldsymbol{\omega}^{(k+1)'} \mathbf{g}^{(k+1)'}, \boldsymbol{\sigma}^{(k+1)'})'$. The above optimization procedure is repeated until convergence; see Algorithm 1.

In practice, we usually specify a maximum number of iteration denoted by $k_{\max}$ for k , and let $\max_{1 \leq j \leq n} |(\theta_j^{(k+1)} - \theta_j^{(k)}) / \theta_j^{(k)}| \leq \delta$ with $\delta > 0$ being the stopping criterion, where $\theta_j^{(k)}$ is the j th element of $\boldsymbol{\theta}^{(k)}$. As a result, the QMLE $\hat{\boldsymbol{\theta}}$ is set to $\boldsymbol{\theta}^{(k^*+1)}$, where $k^* = k_{\max}$ or $\boldsymbol{\theta}^{(k^*+1)}$ satisfies the stopping criterion. Additionally, for the initial point $\boldsymbol{\theta}^{(0)} = (\boldsymbol{\omega}^{(0)'}, \mathbf{g}^{(0)'}, \boldsymbol{\sigma}^{(0)'})'$, note that $\mathbf{g}^{(0)}$ and $\boldsymbol{\sigma}^{(0)}$ can be obtained given $\boldsymbol{\omega}^{(0)}$ by (3.4) and (3.5), respectively. We perform a dense grid search over the space of $\boldsymbol{\omega}$, and select the grid point $\boldsymbol{\omega}$ that minimizes

the loss function of QMLE as the initial point $\boldsymbol{\omega}^{(0)}$.

Algorithm 1 BCD algorithm for QMLE

Input: model order (p, r, s) , initialization $\boldsymbol{\omega}^{(0)}, \boldsymbol{g}^{(0)}$ and $\boldsymbol{\sigma}^{(0)}$.

- 1: **repeat** $k = 0, 1, 2, \dots$
- 2: **for** $i = 1, \dots, r$
- 3: $\lambda_i^{(k+1)} \leftarrow (3.2)$
- 4: **for** $j = 1, \dots, s$
- 5: $\boldsymbol{\eta}_j^{(k+1)} \leftarrow (3.3)$
- 6: $\boldsymbol{g}^{(k+1)} \leftarrow (3.4)$
- 7: $\boldsymbol{\sigma}^{(k+1)} \leftarrow (3.5)$
- 8: Until $k = k_{\max}$ or $\boldsymbol{\theta}^{(k+1)}$ satisfies the stopping criterion.

Output: $\boldsymbol{\theta}^{(k+1)} = (\boldsymbol{\omega}^{(k+1)'}, \boldsymbol{g}^{(k+1)'}, \boldsymbol{\sigma}^{(k+1)'})'$

3.3 Model selection

Section 3.1 introduces the QMLE for model (2.1) under a correctly specified order (p, r, s) . However, the true order is unknown in practice. Now we consider the selection of order (p, r, s) for model (2.1).

In the following, we use $\boldsymbol{\theta}_\iota$ to emphasize their dependence on the order $\iota = (p, r, s)$, and assume the parameter space Θ_ι is a compact subset of $\Omega_\iota \times \mathbb{R}^{N^2 d(\iota)} \times \mathbb{R}^{N(N+1)/2}$ with $\Omega_\iota = \{(-1, 0) \cup (0, 1)\}^r \times \{(0, 1) \times (0, \pi)\}^s$ and $d(\iota) = p + r + 2s$. Let $p_{\max}, r_{\max}$ and $s_{\max}$ be the predetermined positive integers, and $\mathcal{M} = \{\iota \mid 0 \leq p \leq p_{\max}, 0 \leq r \leq r_{\max}, 0 \leq s \leq s_{\max}\}$. In practice, the maximum lags $p_{\max}, r_{\max}$ and $s_{\max}$ are typically specified as small integers (e.g., two or three). This is because, like traditional VARMA models, model (2.1) is inherently parsimonious.

We introduce the following BIC to select the order of model (2.1):

$$\text{BIC}(\iota) = 2T\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_\iota) + n(\iota) \ln T, \quad (3.6)$$

where $\hat{\boldsymbol{\theta}}_\iota$ is the QMLE with the order set to ι , $\tilde{\mathcal{L}}(\hat{\boldsymbol{\theta}}_\iota)$ is the negative Gaussian quasi-log-likelihood evaluated at $\hat{\boldsymbol{\theta}}_\iota$, and $n(\iota) = r + 2s + N^2d(\iota) + N(N+1)/2$ is the dimension of $\hat{\boldsymbol{\theta}}_\iota$; see also Schwarz (1978). Let $\hat{\iota} = \arg \min_{\iota \in \mathcal{M}} \text{BIC}(\iota)$ be the selected order for model (2.1). We prove the selection consistency of BIC under regularity conditions in Section 4.2. Simulation results in Section 5 indicate that BIC performs well in finite samples.

4 Asymptotic results

This section establishes the asymptotic properties of the QMLE $\hat{\boldsymbol{\theta}}$ proposed in Section 3.1 and the selection consistency of the BIC introduced in Section 3.3. Additionally, a comparative analysis is conducted to demonstrate the asymptotic efficiency gain of the proposed QMLE over the least squares estimator (LSE).

4.1 Asymptotic properties for QMLE

Denote $\boldsymbol{\theta}_0 = (\boldsymbol{\alpha}'_0, \boldsymbol{\sigma}'_0)' = (\boldsymbol{\omega}'_0, \mathbf{g}'_0, \boldsymbol{\sigma}'_0)'$ as the true value of $\boldsymbol{\theta}$, where $\boldsymbol{\omega}_0 = (\boldsymbol{\lambda}'_0, \boldsymbol{\eta}'_0)'$ with $\boldsymbol{\lambda}_0 = (\lambda_{10}, \dots, \lambda_{r0})'$ and $\boldsymbol{\eta}_0 = (\boldsymbol{\eta}'_{10}, \dots, \boldsymbol{\eta}'_{s0})'$, $\mathbf{g}_0 = \text{vec}(G_{01}, \dots, G_{0d})$, and $\boldsymbol{\sigma}_0 = \text{vech}(\Sigma_0)$. Let $\boldsymbol{\varepsilon}_{0t} = \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)$. To establish the asymptotic properties for the QMLE $\hat{\boldsymbol{\theta}}$, we introduce the following assumptions.

Assumption 1. (i) The elements of $\boldsymbol{\omega}_0$ are nonzero, where $\lambda_{01}, \dots, \lambda_{0r}$ are distinct, and $\gamma_{01}, \dots, \gamma_{0s}$ are distinct; (ii) there exists a positive constant C such that $\|G_{0k}\| > C$ for $p+1 \leq k \leq d$.

Assumption 2. $\{\mathbf{y}_t, t \in Z\}$ is strictly stationary and ergodic.

Assumption 3. $\{\boldsymbol{\varepsilon}_{0t}, t \in Z\}$ is a sequence of i.i.d. random vectors with $E(\boldsymbol{\varepsilon}_{0t}) = \mathbf{0}_{N \times 1}$ and $E(\boldsymbol{\varepsilon}_{0t}\boldsymbol{\varepsilon}'_{0t}) = \Sigma_0$, where Σ_0 is a positive definite and finite matrix.

Assumption 4. (i) The parameter space Θ is compact and the true parameter vector $\boldsymbol{\theta}_0$ is an interior point in Θ ; (ii) $|\lambda_1|, \dots, |\lambda_r|, \gamma_1, \dots, \gamma_s \in \Lambda$, where Λ is a compact subset of $(0, \rho)$, and $0 < \rho < 1$ is a constant; (iii) $\Sigma(\boldsymbol{\sigma})$ is a finite and positive definite symmetric matrix with its eigenvalue having a positive lower and upper bound denoted by $\underline{\sigma}$ and $\bar{\sigma}$, respectively.

Assumption 1 is required for model identification, ensuring the uniqueness of the true parameter vector $\boldsymbol{\alpha}_0$ for model (2.3) with order (p, r, s) ; see Lemma A5 in the Appendix for details. Specifically, Assumption 1(i) requires $\lambda_{01}, \dots, \lambda_{0r}, \gamma_{01}, \dots, \gamma_{0s}$ to be nonzero and different for identification, and Assumption 1(ii) holds if and only if there exist nonzero elements in G_{0k} . Assumption 1 is also necessary to guarantee the true order (p, r, s) to be irreducible; see Remark 1 in Section 4.2 for details. For the strict stationarity and ergodicity of $\{\mathbf{y}_t\}$ in Assumption 2, Theorem 2.3 in Huang et al. (2025) provides a sufficient condition for model (2.1) by assuming $\{\boldsymbol{\varepsilon}_{0t}\}$ is a strictly stationary and ergodic sequence with $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^2) < \infty$, that is, there exists $0 < \rho < 1$ such that $\max\{|\lambda_{01}|, \dots, |\lambda_{0r}|, \gamma_{01}, \dots, \gamma_{0s}\} \leq \rho$ and $\sum_{k=1}^p \|G_{0k}\|_{\text{op}} + \rho/(1 - \rho) \sum_{k=p+1}^d \|G_{0k}\|_{\text{op}} < 1$. Assumption 3 is necessary for the consistency and asymptotic normality of QMLEs, which together with Assumption 1 guarantees that $\boldsymbol{\theta}_0$ is the unique minimizer of the expected error loss function $E[\mathcal{L}(\boldsymbol{\theta})]$. Note that Assumption 3 only imposes moment conditions on the *i.i.d.* errors $\{\boldsymbol{\varepsilon}_{0t}\}$, which is weaker than the *i.i.d.* sub-Gaussian condition assumed in Huang et al. (2025) and Zheng (2025). Finally, Assumption 4 includes basic conditions for the consistency and asymptotic normality of QMLEs. The compactness of the parameter space Θ is standard for ensuring consistency, while the requirement that $\boldsymbol{\theta}_0$ lies in the interior of Θ is necessary for asymptotic normality.

Let $n = r + 2s + N^2d + N(N + 1)/2$ be the dimension of $\boldsymbol{\theta}$. Define the $n \times n$ matrices

$$\mathcal{I} = E \left(\frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'} \right) \quad \text{and} \quad \mathcal{J} = E \left(\frac{\partial^2 l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right).$$

Theorem 1. Suppose that Assumptions 1–4 hold and $E(\|\mathbf{y}_t\|_2^2) < \infty$. As $T \rightarrow \infty$, we have (i) $\hat{\boldsymbol{\theta}} \rightarrow_p \boldsymbol{\theta}_0$; (ii) furthermore, if $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^4) < \infty$ and $\mathcal{I}$ is positive definite, then $\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(0_{n \times 1}, \mathcal{J}^{-1} \mathcal{I} \mathcal{J}^{-1})$.

Theorem 1 establishes the consistency and asymptotic normality for the QMLE $\hat{\boldsymbol{\theta}}$. We can

see that the consistency requires $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^2) < \infty$, whereas the asymptotic normality further needs $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^4) < \infty$. Moreover, the consistency established in Theorem 1 also relaxes the assumption of sub-Gaussian errors imposed in Huang et al. (2025) and Zheng (2025). If $\boldsymbol{\varepsilon}_{0t}$ is multivariate normal, then the QMLE $\hat{\boldsymbol{\theta}}$ reduces to the MLE and its asymptotic normality can be simplified to $\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(0_{n \times 1}, \mathcal{J}^{-1})$ as $T \rightarrow \infty$.

To calculate the asymptotic variances of $\hat{\boldsymbol{\theta}}$, we can estimate the matrices $\mathcal{I}$ and $\mathcal{J}$ using their sample averages with $\boldsymbol{\theta}_0$ replaced by $\hat{\boldsymbol{\theta}}$, leading to a consistent estimator of the covariance matrix denoted by $\widehat{\text{Var}}(\hat{\boldsymbol{\theta}}) = \hat{\mathcal{J}}^{-1} \hat{\mathcal{I}} \hat{\mathcal{J}}^{-1} / T$. Then we can construct the significance test for each θ_j using the z test statistic $z_j = \hat{\theta}_j / \sqrt{[\widehat{\text{Var}}(\hat{\boldsymbol{\theta}})]_{jj}}$, where θ_j (or $\hat{\theta}_j$) is the j th element of $\boldsymbol{\theta}$ (or $\hat{\boldsymbol{\theta}}$), and $[\widehat{\text{Var}}(\hat{\boldsymbol{\theta}})]_{jj}$ is the j th diagonal entry of $\widehat{\text{Var}}(\hat{\boldsymbol{\theta}})$. Since $z_j \rightarrow_d N(0, 1)$ as $T \rightarrow \infty$, we can conclude that $\theta_j \neq 0$ if $|z_j| > z_{\alpha/2}$ at the significance level α , where z_{α} is the α th quantile of standard normal distribution.

4.2 Selection consistency for BIC

Let $\iota^* = (p^*, r^*, s^*)$ be the true order of model (2.1). Theorem 2 proves that the selected order $\hat{\iota}$ via BIC converges to the true order ι^* consistently under regularity conditions.

Theorem 2. *Suppose ι^* is irreducible. Under the conditions of Theorem 1, if $\iota^* \in \mathcal{M}$, then $P(\hat{\iota} = \iota^*) \rightarrow 1$ as $T \rightarrow \infty$.*

Note that assuming ι^* to be irreducible is necessary for proving the selection consistency of BIC in Theorem 2. If ι^* is reducible, then there exists $\iota^\dagger \neq \iota^*$ such that model (2.1) with order $\iota^\dagger$ generates the same data process as model (2.1) with order ι^* . To avoid this identification issue, we assume that ι^* is irreducible, and provide a sufficient condition for its irreducibility in Remark 1.

Remark 1 (A sufficient condition for irreducible ι^*). *It can be verified that the order (p, r, s) cannot be reduced to smaller ones if Assumption 1 and $G_{0p} \neq \sum_{i=1}^r G_{0,p+i} + \sum_{j=1}^s G_{0,p+r+2j-1}$ hold. Specifically, Assumption 1 ensures the orders r and s non-degenerate, whereas the restriction on $G_{0p} \neq \sum_{i=1}^r G_{0,p+i} + \sum_{j=1}^s G_{0,p+r+2j-1}$ guarantees p non-decreasing; see the proof of this remark in the Appendix for details.*

4.3 Efficiency comparison between QMLE and LSE

For scalable ARMA models in the high-dimensional setting with a large N , [Zheng \(2025\)](#) and [Huang et al. \(2025\)](#) developed the regularized LSEs and derived their non-asymptotic error bounds under the sparsity and low-rank conditions on the parameter matrices G_k for $1 \leq k \leq d$, respectively. For scalable ARMA models with a fixed N , we can also consider the LSE without any sparse or low-rank conditions imposed on G_k 's. This section compares the proposed QMLE in Section 3.1 with the LSE theoretically.

Let $\mathcal{A}$ be the parameter space of $\boldsymbol{\alpha} = (\boldsymbol{\omega}', \boldsymbol{g}')'$, where $\mathcal{A}$ is a subset of $\{(-1, 0) \cup (0, 1)\}^r \times \{(0, 1) \times (0, \pi)\}^s \times \mathbb{R}^{N^2 d}$. Recall that the error function $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\omega}, \boldsymbol{g}) \mathbf{y}_{t-h}$. Denote $\mathcal{L}_{LS}(\boldsymbol{\alpha}) = T^{-1} \sum_{t=1}^T \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\|_2^2$ as the squared error loss function. Similarly, we simply set $\mathbf{y}_s = \mathbf{0}_{N \times 1}$ for $s \leq 0$ and denote the resulting function of $\mathcal{L}_{LS}(\boldsymbol{\alpha})$ as $\tilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha})$. Then the LSE for $\boldsymbol{\alpha}$ is defined as

$$\hat{\boldsymbol{\alpha}}_{LS} = \arg \min_{\boldsymbol{\alpha} \in \mathcal{A}} \tilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha}), \quad (4.1)$$

and the LSE for Σ is $\hat{\Sigma}_{LS} = T^{-1} \sum_{t=1}^T \tilde{\boldsymbol{\varepsilon}}_t(\hat{\boldsymbol{\alpha}}_{LS}) \tilde{\boldsymbol{\varepsilon}}_t'(\hat{\boldsymbol{\alpha}}_{LS})$. Let $n_{\alpha} = r + 2s + N^2 d$ be the dimension of $\boldsymbol{\alpha}$, and define the $n_{\alpha} \times n_{\alpha}$ matrices

$$\mathcal{I}_{LS} = E \left(\frac{\partial \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \Sigma_0 \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} \right) \quad \text{and} \quad \mathcal{J}_{LS} = E \left(\frac{\partial \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} \right).$$

Proposition 1. *Suppose that Assumptions 1–4 hold and $E(\|\mathbf{y}_t\|_2^2) < \infty$. As $T \rightarrow \infty$, we have (i) $\hat{\boldsymbol{\alpha}}_{LS} \rightarrow_p \boldsymbol{\alpha}_0$ and $\hat{\Sigma}_{LS} \rightarrow_p \Sigma_0$; (ii) $\sqrt{T}(\hat{\boldsymbol{\alpha}}_{LS} - \boldsymbol{\alpha}_0) \rightarrow_d N(0_{n_{\alpha} \times 1}, \Xi_{LS})$, where $\Xi_{LS} = \mathcal{J}_{LS}^{-1} \mathcal{I}_{LS} \mathcal{J}_{LS}^{-1}$; (iii) furthermore, if $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^4) < \infty$, then $\sqrt{T}(\text{vec}(\hat{\Sigma}_{LS}) - \text{vec}(\Sigma_0)) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K})$, where $\mathcal{K} = \text{var}[\text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}')]$.*

Proposition 1 establishes the consistency and asymptotic normality for the LSEs $\hat{\boldsymbol{\alpha}}_{LS}$ and $\hat{\Sigma}_{LS}$. It is noteworthy that the consistency of $\hat{\boldsymbol{\alpha}}_{LS}$ and $\hat{\Sigma}_{LS}$ as well as the asymptotic normality of $\hat{\boldsymbol{\alpha}}_{LS}$ only require $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^2) < \infty$, whereas the asymptotic normality of $\hat{\Sigma}_{LS}$ needs $E(\|\boldsymbol{\varepsilon}_{0t}\|_2^4) < \infty$. In comparison with the consistency result in [Huang et al. \(2025\)](#) and [Zheng \(2025\)](#), Proposition 1 relaxes the sub-Gaussian error assumption in their high-

dimensional setting.

To compare the asymptotic efficiency of the proposed QMLE in (3.1) and the LSE in (4.1), we can compare the asymptotic covariance matrices of the estimators for α and Σ separately; see also Poskitt and Salau (1994). Based on Theorem 1, we can prove that $\sqrt{T}(\hat{\alpha} - \alpha_0) \rightarrow_d N(0_{n_\alpha \times 1}, \Xi_{QML})$ and $\sqrt{T}(\text{vec}(\hat{\Sigma}) - \text{vec}(\Sigma_0)) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K})$ as $T \rightarrow \infty$, where $\hat{\sigma} = \text{vech}(\hat{\Sigma})$, and $\Xi_{QML} = \{E[(\partial \epsilon'_t(\alpha_0)/\partial \alpha)\Sigma_0^{-1}(\partial \epsilon_t(\alpha_0)/\partial \alpha')]\}^{-1}$; see details in the proof of Proposition 2 in the Appendix. These, together with Proposition 1, are used to verify the following proposition.

Proposition 2. *Under the conditions of Theorem 1 and Proposition 1, it holds that (i) the asymptotic covariance matrices of the LSE $\hat{\alpha}_{LS}$ and the QMLE $\hat{\alpha}$ satisfy $\Xi_{LS} - \Xi_{QML} \geq 0$, where the equality holds if and only if $\Sigma_0 = \sigma^2 I_N$ for some $\sigma^2 > 0$; (ii) the LSE $\text{vec}(\hat{\Sigma}_{LS})$ and the QMLE $\text{vec}(\hat{\Sigma})$ have the same asymptotic covariance matrix $\mathcal{K} = \text{var}[\text{vec}(\epsilon_{0t}\epsilon'_{0t})]$.*

Proposition 2 indicates that the LSE $\hat{\alpha}_{LS}$ is less efficient than the QMLE $\hat{\alpha}$ when the innovation ϵ_{0t} in model (2.3) has a general covariance matrix, and they are asymptotically equivalent if and only if the innovation ϵ_{0t} is homogeneous and uncorrelated among its components. Moreover, the LSE $\hat{\Sigma}_{LS}$ and the QMLE $\hat{\Sigma}$ are asymptotically equivalent. Thus, we recommend the QMLE in practice owing to its efficiency gain for general situations.

To measure the relative efficiency of $\hat{\alpha}$ and $\hat{\alpha}_{LS}$ numerically, we can compute the asymptotic relative efficiency (ARE) of $\hat{\alpha}_{LS}$ to $\hat{\alpha}$, which is defined as $\text{ARE}(\hat{\alpha}_{LS}, \hat{\alpha}) = (|\Xi_{QML}|/|\Xi_{LS}|)^{1/n_\alpha}$; see Serfling (2009). Given a true parameter α_0 , we can approximate the asymptotic covariance matrices Ξ_{LS} and Ξ_{QML} using their sample averages from a large generated sequence, and then calculate the ARE. Simulation results in Section 5 further verify the theoretical findings that the QMLE $\hat{\alpha}$ is asymptotically more efficient than the LSE $\hat{\alpha}_{LS}$ while the QMLE $\hat{\Sigma}$ is asymptotically equivalent to the LSE $\hat{\Sigma}_{LS}$.

5 Simulation studies

This section conducts simulation experiments to examine the finite sample performance of the proposed estimator QMLE and BIC for order selection. Moreover, we compare the

asymptotic efficiency between the QMLE and the LSE.

The scalable ARMA model (2.1) is proposed as a tractable alternative to the VARMA model (Huang et al., 2025; Zheng, 2025). Thus, we generate data from VARMA models that can be rewritten as scalable ARMA models in (2.1). Specifically, we consider the following VARMA(1, 1) model as the data generating process (DGP):

$$\mathbf{y}_t = \Phi \mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_{0t} - \Theta \boldsymbol{\varepsilon}_{0,t-1}, \quad (5.1)$$

where $\{\boldsymbol{\varepsilon}_{0t}\}$ is a sequence of *i.i.d.* random vectors with zero mean and covariance matrix Σ_0 , and the AR and MA coefficient matrices are specified via Jordan decompositions $\Phi = B_1 J_{\text{AR}} B_1^{-1}$ and $\Theta = B_2 J_{\text{MA}} B_2^{-1}$, respectively. Here B_1 and B_2 are $N \times N$ orthogonal matrices, J_{AR} is a diagonal matrix, and $J_{\text{MA}} = \text{diag}\{\lambda_1, \dots, \lambda_r, C_1, \dots, C_s, \mathbf{0}'_{(N-r-2s) \times 1}\}$ with C_j for $1 \leq j \leq s$ being 2×2 block matrices defined as follows

$$C_j = \gamma_j \begin{pmatrix} \cos(\varphi_j) & \sin(\varphi_j) \\ -\sin(\varphi_j) & \cos(\varphi_j) \end{pmatrix}.$$

Then, this VARMA(1, 1) model can be rewritten into a scalable ARMA model in (2.1) with order $(p, r, s) = (1, r, s)$. Particularly, we set J_{AR} with moduli less than one and generate a random orthogonal matrix B_1 , yielding Φ . For the MA component, we specify λ_i 's and C_j 's to obtain J_{MA} , and get Θ by independently randomly generating orthogonal matrix B_2 . Consequently, the corresponding G_k 's can be obtained as in Zheng (2025); see details in Appendix Section A4.

5.1 QMLE

The first experiment aims to examine the finite sample performance of the QMLE $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\alpha}}', \hat{\boldsymbol{\sigma}}')'$ proposed in Section 3.1.

The data is generated from model (5.1) under the following setting:

$$\text{DGP1} : N = 6, (p, r, s) = (1, 1, 1), (\lambda_{01}, \gamma_{01}, \varphi_{01}) = (-0.8, 0.8, \pi/4),$$

$$J_{\text{AR}} = \text{diag}\{0.5, 0.5, 0.5, -0.5, -0.5, -0.5\}.$$

Moreover, $\{\varepsilon_{0t}\}$ follow the multivariate normal or Student's t_5 distribution with zero mean and covariance matrix $\Sigma_0 = a\mathbf{1}_N\mathbf{1}_N' + (1-a)I_N$, where $a = 0$ or 0.5 . The sample size is set to $T = 500, 750$ or 1000 , and 500 replications are generated for each sample size. We employ BCD algorithm to solve the QMLE, where an evenly spaced grid is used for initialization with the step size being 0.2 for λ_i 's (or γ_j 's) and 0.1π for φ_j 's, and $k_{\max} = 100$ and $\delta = 10^{-6}$ are specified for algorithm convergence.

Table 1 summarizes the biases, empirical standard deviations (ESDs) and asymptotic standard deviations (ASDs) of the QMLE $\hat{\boldsymbol{\theta}}$ under DGP1 with two settings of Σ_0 , respectively. To save space, we only present the estimation results for a subset of the elements from the matrices G_k and Σ_0 . The findings can be summarized as follows. First, as the sample size increases, most of the biases, ESDs and ASDs become smaller, and the ESDs get closer to the corresponding ASDs. This confirms the asymptotic results established in Theorem 1. Second, heavier-tailed $\{\varepsilon_{0t}\}$ lead to larger ESDs and ASDs, especially for the covariance parameters σ_{ij} , which is as expected since the heavier tail of ε_{0t} can make the QMLE less efficient.

5.2 Model selection

The second experiment verifies the selection consistency of the proposed BIC in Section 3.3. The data is generated from model (5.1) under the following setting:

$$\text{DGP2} : N = 6, (p, r, s) = (1, 1, 0), \lambda_{01} \in \{0.7, 0.8, 0.9\},$$

$$J_{\text{AR}} = \text{diag}\{-0.8, 0.8, -0.7, 0.7, -0.6, 0.6\},$$

where $\{\varepsilon_{0t}\}$ is specified as in the first experiment. This design makes DGP2 correspond to the scalable ARMA model (2.1) with the true order $\iota^* = (p^*, r^*, s^*) = (1, 1, 0)$. The BIC in (3.6) is employed to select the order $\iota = (p, r, s)$ with $p_{\max} = r_{\max} = s_{\max} = 2$. The underfitted, correctly selected, and overfitted models by BIC correspond to $\hat{\iota} \in \mathcal{M}_{\text{under}}, \hat{\iota} \in$

$\mathcal{M}_{\text{true}}$, and $\hat{\iota} \in \mathcal{M}_{\text{over}}$, respectively, where $\mathcal{M}_{\text{under}} = \{\iota \in \mathcal{M} \mid p < p^* \text{ or } r < r^* \text{ or } s < s^*\}$, $\mathcal{M}_{\text{true}} = \{\iota \in \mathcal{M} \mid p = p^*, r = r^*, s = s^*\}$, and $\mathcal{M}_{\text{over}} = \{\iota \in \mathcal{M} \mid p \geq p^*, r \geq r^*, s \geq s^*\} \setminus \{\iota^*\}$. We consider the sample sizes $T = 500, 750$ or 1000 , and generate 500 replications for each sample size.

Table 2 provides the percentages of underfitting, correct selection, and overfitting cases by the BIC under DGP2 with two settings of Σ_0 , respectively. It can be seen that BIC performs better when the sample size T increases, which supports the selection consistency of the BIC in Theorem 2. Moreover, the selection accuracy of BIC improves with larger values of λ , which is expected since the larger λ implies stronger signal in order selection.

5.3 Asymptotic efficiency comparison

In the third experiment, we investigate the efficiency gain of the proposed QMLE in (3.1), relative to the LSE in (4.1). Specifically, we calculate the $\text{ARE}(\hat{\alpha}_{LS}, \hat{\alpha})$ defined in Section 4.3, which measures the relative efficiency of $\hat{\alpha}_{LS}$ to $\hat{\alpha}$.

We generate a sequence of sample size $T = 20000$ from model (5.1) under two settings as follows:

$$N = 6, J_{\text{AR}} = \text{diag}\{-0.8, 0.8, -0.7, 0.7, -0.6, 0.6\}$$

$$\text{DGP3(a)} : (p, r, s) = (1, 2, 0), \lambda_{01} = -\lambda_{02} = \lambda \text{ with } \lambda \in \{0.2, 0.4, 0.6, 0.8\};$$

$$\text{DGP3(b)} : (p, r, s) = (1, 0, 1), (\gamma_{01}, \varphi_{01}) = (\gamma, \pi/4) \text{ with } \gamma \in \{0.2, 0.4, 0.6, 0.8\}.$$

Varying λ and γ allows us to examine how relative efficiency of $\hat{\alpha}_{LS}$ to $\hat{\alpha}$ changes with these parameters. Specifically, larger values of λ and γ imply the roots of the MA characteristic equation $|I_N - \Theta B| = 0$ are nearer to the unit circle. To evaluate the impact of Σ_0 on ARE, we let $\{\epsilon_{0t}\}$ follow either a multivariate normal or Student's t_5 distribution with zero mean and different covariance matrices, i.e., $\Sigma_0 = a\mathbf{1}_N\mathbf{1}_N' + (1 - a)I_N$, where $a = 0, 0.6$, or 0.9 . This design allows a to control the degree of cross-sectional correlation, with a larger value of a indicating that Σ_0 deviates further from a diagonal matrix.

Table 3 presents $\text{ARE}(\hat{\alpha}_{LS}, \hat{\alpha})$ under DGP3(a) and DGP3(b) with different settings for

Θ and Σ_0 . It can be observed that the ARE is almost one when $\Sigma_0 = I_N$, which confirms the asymptotic equivalence between the QMLE and LSE for this case established in Proposition 2(i). By contrast, the ARE decreases as the off-diagonal elements of Σ_0 increase. This implies that the concurrent correlation between components of ε_{0t} will lead to efficiency gain of the QMLE relative to the LSE. Moreover, when Σ_0 is not proportional to an identity matrix, the efficiency gain of the QMLE over the LSE becomes slightly more pronounced as λ or γ increases. This observation aligns with the finding for univariate ARMA models in Brockwell and Davis (2009), which states that the efficiency gain of the MLE over the LSE becomes more significant as the roots of the MA characteristic equation move closer to the unit circle.

In summary, the above simulation results confirm the asymptotic results in Section 4. Moreover, the results indicate that the QMLE outperforms the LSE in efficiency for the scalable ARMA model, and thus we suggest to use the QMLE method in practice.

6 An empirical example

This section analyzes six monthly U.S. macroeconomic indicators from the FRED-MD database (McCracken and Ng, 2016), covering the period from January 1959 to August 2025. The indicators include industrial production (IP), the civilian unemployment rate (UR), the consumer price index (CPI), real wages (RW), real personal consumption expenditures (CONS), and the federal funds rate (FFR). This set of indicators is chosen to provide a parsimonious while comprehensive profile of real economic activity, labor market conditions, inflation dynamics, household demand, and monetary policy.

Following McCracken and Ng (2016), each series is transformed to be stationary as summarized in Table 4, and then standardized to have zero mean and unit variance. As a result, we obtain a time series $\{\mathbf{y}_t\}$ of dimension $N = 6$ and length $T = 798$. Figure 1 reports the multivariate autocorrelation matrix (MACM) of the standardized series up to lag 15, which reveals persistent cross-lag dependence with slow decay and occasional oscillatory patterns. Such behavior motivates us to analyze $\mathbf{y}_t$ using the scalable ARMA model (2.1) and the

proposed inference tools in Section 3.

We first fit model (2.1) to the full sample. Using the BIC in Section 3.3 with $p_{\max} = r_{\max} = s_{\max} = 2$, the selected order is $(p, r, s) = (1, 1, 0)$. This together with the representation in (2.2) implies that the dynamics can be decomposed into a short-run autoregressive component and a geometrically decaying infinite-lag component,

$$\mathbf{y}_t = G_1 \mathbf{y}_{t-1} + \sum_{h=1}^{\infty} \lambda_1^h G_2 \mathbf{y}_{t-1-h} + \boldsymbol{\varepsilon}_t. \quad (6.1)$$

The estimated decay parameter is $\hat{\lambda}_1 = 0.84$ with an asymptotic standard error of 0.02, which is highly statistically significant. This strongly supports the presence of a persistent long-run adjustment channel. Additionally, the positive value of λ_1 indicates a gradual and stable long-run response to past shocks, consistent with gradual and persistent adjustment in macroeconomic dynamics over time. Figure 2 shows heatmaps of the estimates for coefficient matrices G_1 and G_2 , with statistical significance indicated by asymptotic standard errors. From model (6.1), G_1 captures short-run dynamics, and its heatmap reveals significant immediate cross-variable transmission. The matrix G_2 governs the long-run dynamics, with its heatmap depicting the pattern of persistent influences. Moreover, Figure 2 shows that Real Wages (RW) and Real Personal Consumption Expenditures (CONS) have immediate effects on most macroeconomic variables. In contrast, Industrial Production (IP), Unemployment Rate (UR), and Federal Funds Rate (FFR) primarily exert long-run influences on the system. Overall, these findings highlight the complex interplay of short- and long-term factors within the macroeconomic system.

To evaluate the forecasting performance of the fitted models, we next conduct one-step-ahead predictions through a rolling forecasting procedure with a fixed moving window. Specifically, we begin with the forecast origin $t_0 = n_0 + 1 = 739$, and fit the scalable ARMA model (2.1) with order $(1, 1, 0)$ by the QMLE method using the data from the beginning to $n_0 = 738$. We then compute the one-month-ahead forecast for $\mathbf{y}_{t_0+1}$, denoted by $\hat{\mathbf{y}}_{t_0+1} = \hat{G}_1 \mathbf{y}_{t_0} + \sum_{h=1}^{\infty} \hat{\lambda}_1^h \hat{G}_2 \mathbf{y}_{t_0-h}$, where $\hat{\lambda}_1$, $\hat{G}_1$ and $\hat{G}_2$ are the QMLEs. After each prediction, we move the window forward by one, repeating the procedure until all data are

used. In total, we obtain $T_{\text{test}} = 60$ one-month-ahead forecasts, using data from the last 5 years as the test set. To assess the forecasting accuracy, we calculate the root mean squared forecast error (RMSFE) and mean absolute forecast error (MAFE) defined as $\text{RMSFE} = [(NT_{\text{test}})^{-1} \sum_{t=t_0+1}^T \|\hat{\mathbf{y}}_t - \mathbf{y}_t\|_2^2]^{1/2}$ and $\text{MAFE} = (NT_{\text{test}})^{-1} \sum_{t=t_0+1}^T \|\hat{\mathbf{y}}_t - \mathbf{y}_t\|_1$, respectively. To facilitate comparison with the VAR benchmark, we further report the relative improvement, defined for method M as $\text{RI}_{\text{RMSFE}}(\text{M}) = \{1 - \text{RMSFE}(\text{M}) / \text{RMSFE}(\text{VAR})\} \times 100\%$ and $\text{RI}_{\text{MAFE}}(\text{M}) = \{1 - \text{MAFE}(\text{M}) / \text{MAFE}(\text{VAR})\} \times 100\%$. A positive (or negative) value indicates that method M yields a lower (or higher) forecast error than the VAR benchmark.

To compare the forecasting performance of the scalable ARMA model fitted by the proposed QMLE method (SARMA_1) with other counterparts, we also perform the rolling forecasting procedure for the following methods:

- SARMA_2 : Fit a scalable ARMA model with order $(p, r, s) = (1, 1, 0)$ to $\{\mathbf{y}_t\}$ using the LSE method;
- VAR: Fit a VAR(2) model to $\{\mathbf{y}_t\}$ using the LSE method, where the order 2 is selected by BIC;
- VARMA_1 : Fit a VARMA(1, 1) model to $\{\mathbf{y}_t\}$ using the QMLE method;
- VARMA_2 : Fit a VARMA(1, 1) model to $\{\mathbf{y}_t\}$ using the iterative ordinary least squares (IOLS) method ([Dias and Kapetanios, 2018](#)).

Since the LSE and QMLE of coefficient matrices are equivalent for the VAR model, we only compare our methods with the VAR model fitted by the least squares method. For comparison of the forecasting performance, Table 5 reports the relative improvements for each method over the VAR benchmark. We conclude that the scalable ARMA model has the smallest RMSFEs and MAFEs, and thus outperforms the VAR and VARMA models in the out-of-sample forecast. Particularly, the scalable ARMA model fitted by the QMLE approach performs the best. These results collectively affirm the advantages of the scalable ARMA framework in modeling and forecasting multivariate time series. As a result, we

suggest the scalable ARMA model fitted by the QMLE method as a competitive and reliable approach for multivariate time series analysis.

7 Conclusion and discussion

This paper develops theoretically grounded inferential tools for the scalable ARMA model, including the quasi-maximum likelihood method for estimation and the BIC method for order selection. By establishing the asymptotic properties of the proposed QMLE, we enable formal significance testing for the scalable ARMA model to uncover causal relationships among time series components. The QMLE is proven to be asymptotically more efficient than the LSE in the presence of cross-sectionally correlated or heteroskedastic innovations. For efficient computation, we also introduce a BCD algorithm. Moreover, the superior efficiency and forecasting accuracy of the QMLE over the LSE are demonstrated via simulations and an application to macroeconomic indicators.

The future research may extend this work in two ways. First, more robust inference methods can be investigated for the scalable ARMA model to relax the moment conditions on the process and innovation. Second, the proposed inference framework can be extended to more general multivariate time series models such as the scalable ARMA model with factors. We leave these topics for future investigation.

Table 1: Biases ($\times 10$), ESDs ($\times 10$) and ASDs ($\times 10$) of the QMLE for DGP1, where $\{\epsilon_{0t}\}$ follow the multivariate normal or Student's t_5 distribution with zero mean and covariance matrix $\Sigma_0 = a\mathbf{1}_N\mathbf{1}'_N + (1 - a)I_N$.

	T	Normal ($a = 0$)			t_5 ($a = 0$)			Normal ($a = 0.5$)			t_5 ($a = 0.5$)		
		Bias	ESD	ASD	Bias	ESD	ASD	Bias	ESD	ASD	Bias	ESD	ASD
λ	500	-0.119	0.728	0.362	-0.111	0.722	0.366	0.071	1.279	0.528	0.109	1.237	0.538
	750	-0.063	0.506	0.323	-0.055	0.541	0.326	0.057	0.855	0.444	0.027	0.900	0.442
	1000	-0.028	0.410	0.293	-0.034	0.401	0.294	0.021	0.623	0.383	0.025	0.651	0.383
γ	500	0.094	0.498	0.260	0.100	0.499	0.261	0.089	0.770	0.381	0.090	0.807	0.383
	750	0.049	0.340	0.227	0.049	0.341	0.228	0.041	0.536	0.333	0.066	0.527	0.331
	1000	0.027	0.283	0.203	0.029	0.264	0.203	0.021	0.405	0.298	0.036	0.424	0.298
ϕ	500	-0.030	0.610	0.313	-0.037	0.599	0.314	-0.074	0.917	0.431	-0.096	0.970	0.432
	750	-0.008	0.401	0.275	-0.018	0.420	0.277	-0.040	0.610	0.374	-0.038	0.610	0.372
	1000	-0.002	0.354	0.248	-0.012	0.352	0.249	-0.037	0.498	0.335	-0.038	0.513	0.335
$g_{1,11}$	500	-0.023	0.454	0.438	-0.036	0.464	0.441	-0.092	0.605	0.568	-0.062	0.603	0.571
	750	-0.015	0.372	0.358	-0.012	0.372	0.360	-0.038	0.495	0.464	-0.016	0.495	0.466
	1000	-0.020	0.329	0.310	-0.024	0.329	0.311	-0.043	0.423	0.402	-0.038	0.424	0.403
$g_{2,11}$	500	0.189	0.764	0.640	0.211	0.773	0.643	0.191	1.220	0.946	0.270	1.207	0.956
	750	0.138	0.587	0.528	0.130	0.591	0.531	0.163	0.858	0.742	0.153	0.925	0.744
	1000	0.094	0.484	0.461	0.093	0.484	0.461	0.094	0.664	0.630	0.099	0.679	0.632
$g_{3,11}$	500	-0.057	0.538	0.499	-0.061	0.525	0.500	-0.038	0.887	0.697	-0.083	0.845	0.702
	750	-0.039	0.413	0.409	-0.027	0.410	0.409	0.001	0.685	0.557	-0.030	0.592	0.545
	1000	-0.034	0.371	0.356	-0.031	0.358	0.356	-0.018	0.519	0.475	-0.038	0.493	0.472
$g_{4,11}$	500	0.058	0.433	0.397	0.077	0.440	0.399	0.099	0.631	0.535	0.079	0.642	0.536
	750	0.046	0.351	0.326	0.034	0.346	0.327	0.082	0.470	0.426	0.022	0.475	0.422
	1000	0.039	0.294	0.284	0.033	0.299	0.284	0.071	0.387	0.364	0.035	0.394	0.364
σ_{11}	500	-0.495	0.580	0.597	-0.487	1.090	0.973	-0.523	0.595	0.596	-0.522	1.088	0.966
	750	-0.327	0.486	0.497	-0.294	0.956	0.834	-0.331	0.521	0.497	-0.309	0.952	0.831
	1000	-0.254	0.426	0.433	-0.295	0.749	0.732	-0.261	0.450	0.433	-0.305	0.745	0.730
σ_{21}	500	0.010	0.436	0.424	0.015	0.691	0.616	-0.310	0.466	0.471	-0.282	0.807	0.706
	750	-0.003	0.352	0.353	0.012	0.635	0.546	-0.197	0.390	0.392	-0.163	0.768	0.620
	1000	0.003	0.315	0.308	0.033	0.491	0.478	-0.155	0.333	0.343	-0.135	0.540	0.550
σ_{31}	500	-0.048	0.450	0.600	-0.087	0.700	0.955	-0.317	0.466	0.596	-0.371	0.798	0.952
	750	-0.027	0.358	0.500	-0.000	0.610	0.849	-0.195	0.389	0.496	-0.175	0.773	0.856
	1000	-0.015	0.315	0.437	-0.017	0.501	0.771	-0.146	0.335	0.436	-0.170	0.581	0.767

Table 2: Percentages of underfitted, correctly selected and overfitted cases under DGP2, where the BIC is based on the QMLE, and $\boldsymbol{\varepsilon}_{0t}$ follow the multivariate normal or Student's t_5 distribution with zero mean and covariance matrix $\Sigma_0 = a\mathbf{1}_N\mathbf{1}'_N + (1 - a)I_N$.

λ	T	Normal ($a = 0$)			t_5 ($a = 0$)			Normal ($a = 0.5$)			t_5 ($a = 0.5$)		
		Under	Exact	Over	Under	Exact	Over	Under	Exact	Over	Under	Exact	Over
0.7	500	95.0	5.0	0.0	93.0	7.0	0.0	92.0	8.0	0.0	89.6	10.4	0.0
0.7	750	18.8	81.2	0.0	18.6	81.4	0.0	11.8	88.2	0.0	8.2	91.8	0.0
0.7	1000	0.2	99.8	0.0	0.2	99.8	0.0	0.0	100.0	0.0	0.2	99.8	0.0
0.8	500	45.2	54.8	0.0	43.2	56.8	0.0	33.6	66.4	0.0	31.8	68.2	0.0
0.8	750	0.4	99.6	0.0	0.4	99.6	0.0	0.2	99.8	0.0	0.2	99.8	0.0
0.8	1000	0.2	99.8	0.0	0.0	100.0	0.0	0.0	100.0	0.0	0.2	99.8	0.0
0.9	500	7.2	92.8	0.0	8.0	92.0	0.0	5.8	94.2	0.0	4.4	95.6	0.0
0.9	750	0.4	99.6	0.0	0.4	99.6	0.0	0.2	99.8	0.0	0.2	99.8	0.0
0.9	1000	0.2	99.8	0.0	0.0	100.0	0.0	0.0	100.0	0.0	0.0	100.0	0.0

Table 3: $\text{ARE}(\hat{\boldsymbol{\alpha}}_{LS}, \hat{\boldsymbol{\alpha}})$ under DGP3(a) and DGP3(b), where $\{\boldsymbol{\varepsilon}_{0t}\}$ follow the multivariate normal or Student's t_5 distribution with zero mean and covariance matrix $\Sigma_0 = a\mathbf{1}_N\mathbf{1}'_N + (1 - a)I_N$.

	a	λ				γ			
		0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
Normal	0.0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	0.6	0.983	0.984	0.983	0.983	0.984	0.984	0.983	0.983
	0.9	0.961	0.960	0.958	0.957	0.961	0.961	0.958	0.956
t_5	0.0	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	0.6	0.983	0.984	0.983	0.983	0.984	0.984	0.983	0.983
	0.9	0.961	0.960	0.958	0.957	0.961	0.961	0.958	0.956

Table 4: Description and transformation of the six macroeconomic indicators, where transformations are coded as : 1 = first difference, 2 = first difference of the log series, and 3 = second difference of the log series.

Variable	Description	Transformation
IP	Industrial Production Index	2
UR	Civilian Unemployment Rate	1
CPI	Consumer Price Index	3
RW	Real wages	2
CONS	Real Personal Consumption Expenditures	2
FFR	Federal Funds Rate	1

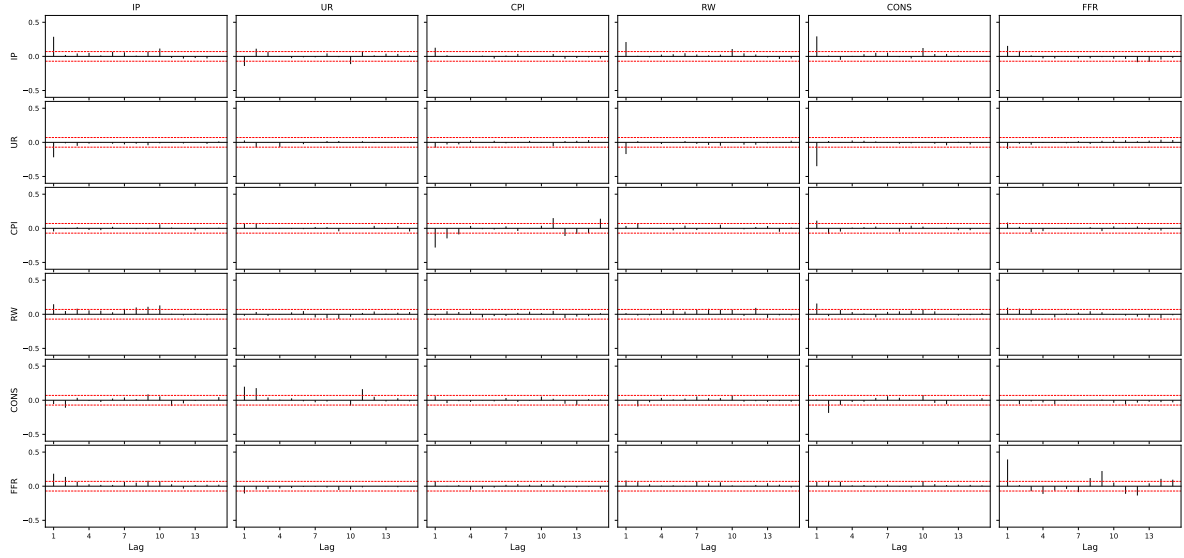


Figure 1: The MACM plots of $\{y_t\}$ with the 95% confidence intervals in dashed lines.

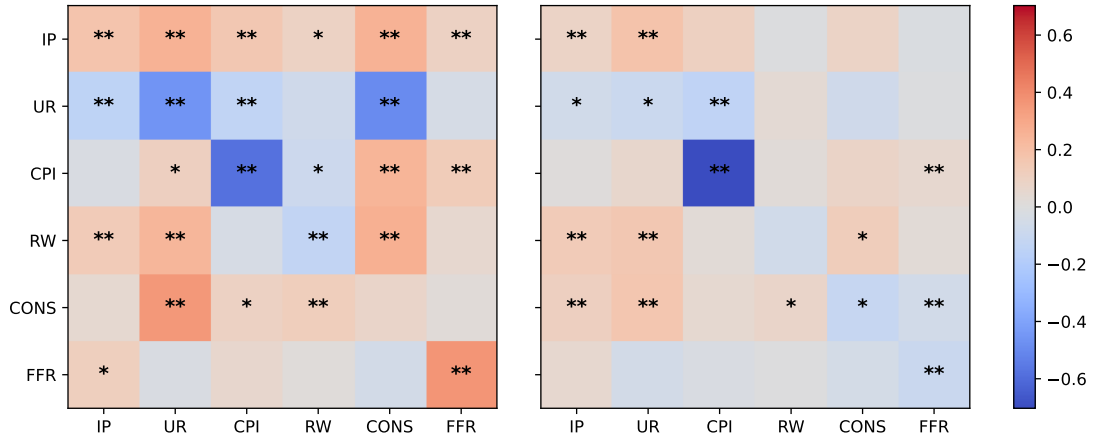


Figure 2: Heatmap of the estimates for coefficient matrices G_1 (left) and G_2 (right) in model (6.1), respectively. Statistically significant coefficients are marked with asterisks: ** for significance at the 1% level and * for significance at the 5% level. The color scale indicates the magnitude of the coefficients.

Table 5: Relative improvement in out-of-sample prediction accuracy over the VAR benchmark for six U.S. macroeconomic indicators. SARMA₁ (or SARMA₂) represents the scalable ARMA model fitted by the QMLE (or LSE) method and VARMA₁ (or VARMA₂) represents the VARMA model fitted by the QMLE (or IOLS) method. The largest improvement in each row is marked in bold.

	SARMA ₁	SARMA ₂	VARMA ₁	VARMA ₂
RI-RMSFE (%)	7.24	6.72	5.94	-6.98
RI-MAFE (%)	3.20	2.85	1.25	-2.85

Appendix

This Appendix provides technical details for Theorems 1–2, Propositions 1–2 and Remark 1, and introduces Lemmas A1–A14 which give some preliminary results for proving Theorem 1 and Proposition 1. It also includes the details on the connection between scalable ARMA models with vector ARMA models. Throughout the Appendix, we denote vectors by small boldface letters $\mathbf{x}, \mathbf{y}, \dots$ and matrices by capital letters $A, B, \dots$. For a vector $\mathbf{x}$, $\|\mathbf{x}\|_p$ denotes its ℓ_p norm (i.e., $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$). For a matrix $A \in \mathbb{R}^{d_1 \times d_2}$, denote by $\|A\|_{\text{op}}$ its operator norm (i.e., $\|A\|_{\text{op}} = \sup_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|$), by $\|A\|$ its sub-multiplicative matrix norm (i.e., $\|AB\| \leq \|A\|\|B\|$ for any matrices A and B), by A' its transpose, and by $\text{vec}(A)$ its vectorization. For a square matrix B , $|B|$, $\text{tr}(B)$ and $\text{vech}(B)$ are the determinant, trace, and the lower triangular-stacking vector of B , respectively. For square matrices B_j 's, $\text{diag}\{B_1, \dots, B_m\}$ denotes the block diagonal matrix whose main diagonal consists of $B_1, \dots, B_m$. Denote $\otimes$ as the Kronecker product. Moreover, $\mathbf{1}_N$ is the N -dimensional vector with all elements being one, $0_{N \times T}$ is the $N \times T$ zero matrix and I_N is the $N \times N$ identity matrix. In addition, $O(1)$ denotes a bounded series of nonstochastic variables, $O_p(1)$ denotes a sequence of random variables bounded in probability, $o_p(1)$ denotes a sequence of random variables converging to zero in probability, $\rightarrow_p$ denotes the convergence in probability and $\rightarrow_d$ denotes the convergence in distribution.

A1 Lemmas

Lemma A1. *Under Assumption 4, there exists a constant $C > 0$ such that*

$$\begin{aligned} (i) \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\| &\leq C\xi_{\rho,t-1} + \|\boldsymbol{\varepsilon}_{0t}\|; & (ii) \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}} \right\| &\leq C\xi_{\rho,t-1}; \\ (iii) \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial(\text{vec}(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})/\partial \boldsymbol{\alpha}))}{\partial \boldsymbol{\alpha}'} \right\| &\leq C\xi_{\rho,t-p-1}, \end{aligned}$$

where $\xi_{\rho t} = 1 + \sum_{h=0}^{\infty} \rho^h \|\mathbf{y}_{t-h}\|$ with a constant $\rho \in (0, 1)$.

Proof. Recall that $A_h(\boldsymbol{\alpha}) = G_h$ for $1 \leq h \leq p$ and $A_h(\boldsymbol{\alpha}) = \sum_{k=p+1}^d \ell_{hk}(\boldsymbol{\omega}) G_k$ for $j \geq p+1$ by (2.1). Then by Assumption 4, there exists a positive constant C such that $\|G_k\| < C$ for $1 \leq k \leq d$, and then we can obtain that $\|A_h(\boldsymbol{\alpha})\| = \|G_h\| \leq C$ for $1 \leq h \leq p$, and $\|A_h(\boldsymbol{\alpha})\| = \sum_{k=p+1}^d |\ell_{hk}(\boldsymbol{\omega})| \|G_k\| \leq (r+2s)C\rho^{h-p}$ for $h \geq p+1$. Let $C^* = C(r+2s)\rho^{-p}$, it follows that

$$\|A_h(\boldsymbol{\alpha})\| \leq C^* \rho^h \text{ for } h \geq 1. \quad (\text{A1.1})$$

We first show (i). Recall that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$ and $\boldsymbol{\varepsilon}_{0t} = \mathbf{y}_t - \sum_{h=1}^{\infty} A_{0h} \mathbf{y}_{t-h}$. Hence it holds that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t} = \sum_{h=1}^{\infty} [A_{0h} - A_h(\boldsymbol{\alpha})] \mathbf{y}_{t-h}$. Moreover, by Assumption 4(i) and (A1.1), it can be shown that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \|A_{0h} - A_h(\boldsymbol{\alpha})\| \leq 2 \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \|A_h(\boldsymbol{\alpha})\| = O(\rho^h).$$

It then follows that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\| \leq \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t}\| + \|\boldsymbol{\varepsilon}_{0t}\| \leq \sum_{h=1}^{\infty} O(\rho^h) \|\mathbf{y}_{t-h}\| + \|\boldsymbol{\varepsilon}_{0t}\| \leq C\xi_{\rho,t-1} + \|\boldsymbol{\varepsilon}_{0t}\|$$

holds for some constant $C > 0$. Thus (i) holds.

Next we show (ii). Suppose that $\rho_1 < \rho < 1$, it can be easily shown that $h(\rho_1/\rho)^{h-1} < C$

holds for $h \geq 1$ and some constant C . It follows that

$$1 + \sum_{h=1}^{\infty} h \rho_1^{h-1} \|\mathbf{y}_{t-h}\| \leq 1 + \sum_{h=1}^{\infty} h \left(\frac{\rho_1}{\rho} \right)^{h-1} \rho^{h-1} \|\mathbf{y}_{t-h}\| < C \xi_{\rho, t-1}.$$

This together with (A3.1) and Assumption 4, implies that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i} \right\| \leq O(1) \xi_{\rho, t-p-1}, \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\eta}'_j} \right\| \leq O(1) \xi_{\rho, t-p-1} \text{ and } \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \mathbf{g}'} \right\| \leq O(1) \xi_{\rho, t-1},$$

where $1 \leq i \leq r$ and $1 \leq j \leq s$. Then it can be shown that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} \right\| \leq \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left(\sum_{i=1}^r \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i} \right\| + \sum_{j=1}^s \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\eta}'_j} \right\| + \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \mathbf{g}'} \right\| \right) \leq O(1) \xi_{\rho, t-1} \quad (\text{A1.2})$$

Thus (ii) holds.

Similarly, for (iii), suppose that $\rho_1 < \rho < 1$, it can be shown that $h(h-1)(\rho_1/\rho)^{h-2} < C$ holds for $h \geq 1$ and some constant C , and it follows that

$$1 + \sum_{h=2}^{\infty} h(h-1) \rho_1^{h-2} \|\mathbf{y}_{t-h}\| \leq 1 + \sum_{h=2}^{\infty} h(h-1) \left(\frac{\rho_1}{\rho} \right)^{h-2} \rho^{h-2} \|\mathbf{y}_{t-h}\| < C \xi_{\rho, t-2}.$$

This together with (A3.2) and Assumption 4, implies that

$$\begin{aligned} \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i^2} \right\| &\leq O(1) \xi_{\rho, t-p-2}, \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j^2} \right\| \leq O(1) \xi_{\rho, t-p-2}, \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j^2} \right\| \leq O(1) \xi_{\rho, t-p-1}, \\ \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \varphi_j} \right\| &\leq O(1) \xi_{\rho, t-p-1}, \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i \partial \mathbf{g}'_{p+i}} \right\| \leq O(1) \xi_{\rho, t-p-1}, \\ \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \mathbf{g}'_{p+r+2j-1}} \right\| &\leq O(1) \xi_{\rho, t-p-1}, \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \mathbf{g}'_{p+r+2j}} \right\| \leq O(1) \xi_{\rho, t-p-1}, \\ \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j \partial \mathbf{g}'_{p+r+2j-1}} \right\| &\leq O(1) \xi_{\rho, t-p-1} \text{ and } \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j \partial \mathbf{g}'_{p+r+2j}} \right\| \leq O(1) \xi_{\rho, t-p-1}, \end{aligned}$$

where $1 \leq i \leq r$ and $1 \leq j \leq s$. Then with similar argument of (A1.2), we can show that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial(\text{vec}(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})/\partial \boldsymbol{\alpha}))}{\partial \boldsymbol{\alpha}'} \right\| \leq O(1) \xi_{\rho, t-p-1}.$$

As a result, (iii) holds. $\square$

Lemma A2. *Under Assumptions 1–4, if $E(\|\mathbf{y}_t\|^2) < \infty$, it holds that*

$$(i) \ E \left(\sup_{\boldsymbol{\theta} \in \Theta} |l_t(\boldsymbol{\theta})| \right) < \infty; \quad (ii) \ E \left(\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right\| \right) < \infty; \quad (iii) \ E \left(\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right\| \right) < \infty.$$

Proof. We first establish (i). Recall that $l_t(\boldsymbol{\theta}) = 1/2 \ln |\Sigma(\boldsymbol{\sigma})| + 1/2 \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$. It holds that

$$E \left(\sup_{\boldsymbol{\theta} \in \Theta} |l_t(\boldsymbol{\theta})| \right) \leq \frac{1}{2} \sup_{\boldsymbol{\theta} \in \Theta} |\ln |\Sigma(\boldsymbol{\sigma})|| + \frac{1}{2} E \left(\sup_{\boldsymbol{\theta} \in \Theta} \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \right).$$

Thus it suffices to show that (a) $\sup_{\boldsymbol{\theta} \in \Theta} |\ln |\Sigma(\boldsymbol{\sigma})|| < \infty$, and (b) $E(\sup_{\boldsymbol{\theta} \in \Theta} \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})) < \infty$. For (a), under Assumption 4(ii), it can be shown that

$$\begin{aligned} \sup_{\boldsymbol{\theta} \in \Theta} |\ln |\Sigma(\boldsymbol{\sigma})|| &\leq - \sup_{\boldsymbol{\theta} \in \Theta} \ln |\Sigma(\boldsymbol{\sigma})| I(|\Sigma(\boldsymbol{\sigma})| < 1) + \sup_{\boldsymbol{\theta} \in \Theta} \ln |\Sigma(\boldsymbol{\sigma})| I(|\Sigma(\boldsymbol{\sigma})| > 1) \\ &\leq -N \ln(\min\{\underline{\sigma}, 1\}) + N \ln(\max\{\bar{\sigma}, 1\}) < \infty. \end{aligned}$$

For (b), by Assumption 4(ii) and Lemma A1(i), we have that

$$\begin{aligned} E \left(\sup_{\boldsymbol{\theta} \in \Theta} \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \right) &= E \left(\sup_{\boldsymbol{\theta} \in \Theta} \|\boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\| \right) \\ &\leq E \left(\sup_{\boldsymbol{\theta} \in \Theta} \|\Sigma^{-1}(\boldsymbol{\sigma})\| \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\|^2 \right) \\ &= O(1) E \left(\sup_{\boldsymbol{\theta} \in \Theta} \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\|^2 \right) \leq O(1) E(\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|)^2, \end{aligned}$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$. This together with $E(\xi_{\rho t}^2) < \infty$ and $E(\|\boldsymbol{\varepsilon}_{0t}\|^2) < \infty$ by $E(\|\mathbf{y}_t\|^2) < \infty$ and Assumptions 2–3, implies that (b) holds. Thus, (i) holds by (a) and (b).

Next we show (ii). Similarly, by (A3.3), Lemma A1 and Assumption 4, it holds that

$$\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right\| \leq O(1) \sup_{\boldsymbol{\theta} \in \Theta} \left(\left\| \frac{\partial \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}} \right\| \|\Sigma^{-1}(\boldsymbol{\sigma})\| \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\| + \|\Sigma^{-1}(\boldsymbol{\sigma})\| + \|\Sigma^{-1}(\boldsymbol{\sigma})\|^2 \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})\|^2 \right)$$

$$\begin{aligned}
&\leq O(1)\xi_{\rho,t-1}(\xi_{\rho,t-1} + \|\boldsymbol{\varepsilon}_{0t}\|) + O(1)[1 + (\xi_{\rho,t-1} + \|\boldsymbol{\varepsilon}_{0t}\|)^2] \\
&\leq O(1)(\xi_{\rho,t-1}^2 + \|\boldsymbol{\varepsilon}_{0t}\|^2),
\end{aligned}$$

and then $E\left(\sup_{\boldsymbol{\theta} \in \Theta} \|\partial l_t(\boldsymbol{\theta})/\partial \boldsymbol{\theta}\|\right) < \infty$ by $E(\|\boldsymbol{\varepsilon}_{0t}\|^2) < \infty$ and $E(\xi_{\rho t}^2) < \infty$. Thus (ii) holds.

Lastly we prove (iii). With similar argument in the proof of (i) and (ii), (A3.4) and Lemma A1 imply that

$$\begin{aligned}
\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}'} \right\| &\leq O(1)(\xi_{\rho,t-1}^2 + \|\boldsymbol{\varepsilon}_{0t}\|^2), \\
\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\sigma}'} \right\| &\leq O(1)(\xi_{\rho,t-1}^2 + \|\boldsymbol{\varepsilon}_{0t}\|^2), \\
\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}'} \right\| &\leq O(1)(\xi_{\rho,t-1}^2 + \|\boldsymbol{\varepsilon}_{0t}\|^2),
\end{aligned}$$

and then it holds that

$$\begin{aligned}
E\left(\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right\|\right) &\leq E\left[\sup_{\boldsymbol{\theta} \in \Theta} \left(\left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}'} \right\| + \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\sigma}'} \right\| + \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}'} \right\| \right)\right] \\
&\leq O(1)E(\xi_{\rho,t-1}^2 + \|\boldsymbol{\varepsilon}_{0t}\|^2) < \infty.
\end{aligned}$$

Thus (iii) holds. □

Lemma A3. *Under Assumptions 2-4, the following results hold:*

$$\begin{aligned}
(i) \quad &\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{1}{T} \sum_{t=1}^T \{l_t(\boldsymbol{\theta}) - E[l_t(\boldsymbol{\theta})]\} \right| = o_p(1); \\
(ii) \quad &\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} - E\left[\frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] \right\} \right\| = o_p(1); \\
(iii) \quad &\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} - E\left[\frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right] \right\} \right\| = o_p(1).
\end{aligned}$$

Proof. This lemma directly follows from Lemma A2 and Theorem 3.1 of Ling and McAleer (2003), so its proof is omitted here. □

Lemma A4. Under Assumption 4, the following results hold:

$$\begin{aligned}
(i) \quad & \sup_{\boldsymbol{\theta} \in \Theta} |l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|); \\
(ii) \quad & \sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} - \frac{\partial \tilde{l}_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right\| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|); \\
(iii) \quad & \sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} - \frac{\partial^2 \tilde{l}_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right\| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|),
\end{aligned}$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$.

Proof. We first show (i). Recall that $l_t(\boldsymbol{\theta}) = 1/2 \ln |\Sigma(\boldsymbol{\sigma})| + 1/2 \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$. Under Assumption 4(ii), it holds that

$$\begin{aligned}
\sup_{\boldsymbol{\theta}} |l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})| &= \sup_{\boldsymbol{\theta} \in \Theta} |\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})| \\
&\leq O(1) \sup_{\boldsymbol{\theta} \in \Theta} |\tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) (\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}))| \\
&\leq O(1) \sup_{\boldsymbol{\theta} \in \Theta} \|\tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})\| \sup_{\boldsymbol{\theta} \in \Theta} \|\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})\|.
\end{aligned}$$

This together with Lemma A1(i) and (A1.10), implies that

$$\sup_{\boldsymbol{\theta}} |l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|).$$

Hence (i) holds.

Similarly, by (A1.11), (A3.3)–(A3.4) and Assumption 4(ii), we can obtain that (ii) and (iii) hold. $\square$

Lemma A5. Under Assumptions 1 and 3, if $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \boldsymbol{\varepsilon}_{0t}$ a.s., then $\boldsymbol{\alpha} = \boldsymbol{\alpha}_0$.

Proof. Recall that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$ and $\boldsymbol{\varepsilon}_{0t} = \mathbf{y}_t - \sum_{h=1}^{\infty} A_{0h} \mathbf{y}_{t-h}$. It holds that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t} = \sum_{h=1}^{\infty} [A_{0h} - A_h(\boldsymbol{\alpha})] \mathbf{y}_{t-h}$. Thus $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \boldsymbol{\varepsilon}_{0t}$ a.s. implies that

$$\sum_{h=1}^{\infty} (A_h(\boldsymbol{\alpha}) - A_{0h}) \mathbf{y}_{t-h} = 0 \quad \text{a.s..} \tag{A1.3}$$

We first show that $A_h(\boldsymbol{\alpha}) = A_{0h}$ for all $h \geq 1$ by contradiction. Suppose that there exist k 's such that $A_k(\boldsymbol{\alpha}) \neq A_{0k}$. Note that it is obviously impossible that there is only one k such that $A_k(\boldsymbol{\alpha}) \neq A_{0k}$ when $\mathbf{y}_{t-k} \neq 0_{N \times 1}$. Then (A1.3) implies that $\mathbf{y}_t$ can be represented as a linear combination of $\mathbf{y}_{t-k}$'s a.s. And it follows that $\boldsymbol{\varepsilon}_{0t} = \mathbf{y}_t - E(\mathbf{y}_t \mid \mathcal{F}_{t-1}) = 0_{N \times 1}$ a.s., which is in contradiction with that $\text{var}(\boldsymbol{\varepsilon}_{0t}) = \Sigma_0$ is positive definite under Assumption 3. Thus we have that $A_h(\boldsymbol{\alpha}) = A_{0h}$ for all $h \geq 1$.

Recall that $A_h(\boldsymbol{\alpha}) = G_h$ for $1 \leq h \leq p$, and $A_h(\boldsymbol{\alpha}) = \sum_{k=p+1}^d \ell_{hk}(\boldsymbol{\omega}) G_k$ for $h \geq p+1$ with $d = p+r+2s$. Since $A_h(\boldsymbol{\alpha}) = A_{0h}$ for all $h \geq 1$, we have that $G_h = G_{0h}$ for $1 \leq h \leq p$ and

$$\begin{aligned} & \sum_{i=1}^r \lambda_i^m G_{p+i} + \sum_{k=1}^s \gamma_k^m [\cos(m\varphi_k) G_{p+r+2k-1} + \sin(m\varphi_k) G_{p+r+2k}] \\ &= \sum_{i=1}^r \lambda_{0i}^m G_{0,p+i} + \sum_{k=1}^s \gamma_{0k}^m [\cos(m\varphi_{0k}) G_{0,p+r+2k-1} + \sin(m\varphi_{0k}) G_{0,p+r+2k}] \end{aligned} \quad (\text{A1.4})$$

for $m = h - p \geq 1$. Then to prove $\boldsymbol{\alpha} = \boldsymbol{\alpha}_0$ under Assumption 1, it suffices to show that

- (i) Equation (A1.5) below holds for all $m \geq 1$ if and only if $\mathbf{c} = (c_1, \dots, c_r, c_{11}, c_{12}, \dots, c_{s1}, c_{s2})' = 0_{(r+2s) \times 1}$:

$$\sum_{i=1}^r c_i \lambda_i^m + \sum_{k=1}^s [c_{k1} \gamma_k^m \cos(m\varphi_k) + c_{k2} \gamma_k^m \sin(m\varphi_k)] = 0, \quad (\text{A1.5})$$

where $0 < |\lambda_i| < 1$ and $0 < \gamma_k < 1$, and $\{\lambda_i\}$ and $\{\gamma_k\}$ are distinct, respectively;

- (ii) Equation (A1.6) below holds for all $m \geq 1$ if and only if $\varphi_k = \varphi_{0k}$, $G_{p+r+2k-1} = G_{0,p+r+2k-1}$ and $G_{p+r+2k} = G_{0,p+r+2k}$:

$$\cos(m\varphi_k) G_{p+r+2k-1} + \sin(m\varphi_k) G_{p+r+2k} = \cos(m\varphi_{0k}) G_{0,p+r+2k-1} + \sin(m\varphi_{0k}) G_{0,p+r+2k} \quad (\text{A1.6})$$

where $\varphi_k, \varphi_{0k} \in (0, \pi)$.

Since that (A1.4) and (i) imply that $\lambda_i = \lambda_{0i}$ and $G_{p+i} = G_{0,p+i}$ for $1 \leq i \leq r$ as well as

$\gamma_k = \gamma_{0k}$ for $1 \leq k \leq s$, then (A1.6) holds which together with (ii) implies that $\varphi_k = \varphi_{0k}$, $G_{p+r+2k-1} = G_{0,p+r+2k-1}$ and $G_{p+r+2k} = G_{0,p+r+2k}$ for $1 \leq k \leq s$.

For (i), obviously we only need to show that (A1.5) holds for all $m \geq 1$ implies that $\mathbf{c} = 0$. Without loss of generality, we assume that $|\lambda_1| > \dots > |\lambda_r|$ and $\gamma_1 > \dots > \gamma_s$. Let $\tau = \max\{|\lambda_1|, \gamma_1\}$. Dividing (A1.5) by τ^m and considering the dominant terms, we obtain

$$c_1 \left(\frac{\lambda_1}{\tau} \right)^m + c_{11} \left(\frac{\gamma_1}{\tau} \right)^m \cos(m\varphi_1) + c_{12} \left(\frac{\gamma_1}{\tau} \right)^m \sin(m\varphi_1) = 0. \quad (\text{A1.7})$$

Then we discuss (A1.7) in situations below:

- (a) If $\tau = |\lambda_1| = \gamma_1$, then $|c_1| = |c_{11} \cos(m\varphi_1) + c_{12} \sin(m\varphi_1)| = \left| \sqrt{c_{11}^2 + c_{12}^2} \cos(m\varphi_1 - \arccos(\iota)) \right|$ with $\iota = c_{11}/(\sqrt{c_{11}^2 + c_{12}^2})$, and it follows that $c_1 = c_{11} = c_{12} = 0$;
- (b) if $\tau = |\lambda_1| > \gamma_1$, then $c_1 = 0$, and it follows that $c_{11} = c_{12} = 0$ by $c_{11} \cos(m\varphi_1) + c_{12} \sin(m\varphi_1) = 0$;
- (c) if $\tau = \gamma_1 > |\lambda_1|$, then $c_{11} = c_{12} = 0$, and it follows that $c_1 = 0$.

Thus (A1.7) implies that $c_1 = c_{11} = c_{12} = 0$. Similarly it can be shown that the other elements of $\mathbf{c}$ are zeros. It follows that (i) holds.

For (ii), using $a \cos(x) + b \sin(x) = \sqrt{a^2 + b^2} \cos(x - y)$ with $y = \arccos(a/\sqrt{a^2 + b^2})$, it can be easily shown that (A1.6) holds for all $m \geq 1$ if and only if $\varphi_k = \varphi_{0k}$, $G_{p+r+2k-1} = G_{0,p+r+2k-1}$ and $G_{p+r+2k} = G_{0,p+r+2k}$. As a result, $\boldsymbol{\alpha} = \boldsymbol{\alpha}_0$ holds by (i) and (ii). $\square$

Lemma A6. Under Assumptions 1, 3 and 4, $E[l_t(\boldsymbol{\theta})]$ has a unique minimum at $\boldsymbol{\theta}_0$.

Proof. Recall that $l_t(\boldsymbol{\theta}) = 1/2 \ln |\Sigma(\boldsymbol{\sigma})| + 1/2 \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ with $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$, and $\boldsymbol{\varepsilon}_{0t} = \mathbf{y}_t - \sum_{h=1}^{\infty} A_{0h} \mathbf{y}_{t-h}$. It holds that

$$\begin{aligned} 2E[l_t(\boldsymbol{\theta})] &= \ln |\Sigma(\boldsymbol{\sigma})| + E[\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] \\ &= \ln |\Sigma(\boldsymbol{\sigma})| + E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})' \Sigma^{-1}(\boldsymbol{\sigma}) (\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})] \\ &\quad + E[\boldsymbol{\varepsilon}_{0t}' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_{0t}] + 2E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_{0t}]. \end{aligned}$$

Note that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t} = -\sum_{h=1}^{\infty} (A_h(\boldsymbol{\alpha}) - A_{0h}) \mathbf{y}_{t-h} \in \mathcal{F}_{t-1}$ with $\mathcal{F}_t$ being the σ -field generated by $\{\boldsymbol{\varepsilon}_{0s}, s \leq t\}$, and $E(\boldsymbol{\varepsilon}_{0t} \mid \mathcal{F}_{t-1}) = 0$ under Assumption 3. Then using the law of iterated expectation, we have that

$$\begin{aligned} 2E[l_t(\boldsymbol{\theta})] &= \{E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})' \Sigma^{-1}(\boldsymbol{\sigma})(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})]\} + \{\ln |\Sigma(\boldsymbol{\sigma})| + E[\boldsymbol{\varepsilon}_{0t}' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_{0t}]\} \\ &:= Q_1(\boldsymbol{\theta}) + Q_2(\boldsymbol{\sigma}). \end{aligned}$$

By Lemma A5, it can be easily shown that $Q_1(\boldsymbol{\theta})$ has a unique minimum at $\boldsymbol{\theta}_0$. Thus we are left to show that $Q_2(\boldsymbol{\sigma})$ has a unique minimum at $\boldsymbol{\sigma}_0$. Rewrite $Q_2(\boldsymbol{\sigma})$ as follows,

$$Q_2(\boldsymbol{\sigma}) = -\ln |\Sigma_0 \Sigma^{-1}(\boldsymbol{\sigma})| + \text{tr}(\Sigma_0 \Sigma^{-1}(\boldsymbol{\sigma})) + \ln |\Sigma_0|.$$

By Lemma A.6 of Johansen et al. (1995), $-\ln |M| + \text{tr} M \geq N$ holds for any N -dimensional positive definite matrix M , and the equality holds if and only if M is an identity matrix. Note that under Assumptions 3 and 4, $\Sigma_0 \Sigma^{-1}(\boldsymbol{\sigma})$ is positive definite. Thus $Q_2(\boldsymbol{\sigma}) \geq N + \ln |\Sigma_0|$ and the equality holds if and only if $\Sigma_0 \Sigma^{-1}(\boldsymbol{\sigma}) = I_N$, which implies that $Q_2(\boldsymbol{\sigma})$ has a unique minimum at $\boldsymbol{\sigma}_0$. As a result, $E[l_t(\boldsymbol{\theta})]$ is uniquely minimized at $\boldsymbol{\theta}_0$. $\square$

Lemma A7. *Under Assumptions 1 and 3, $\mathcal{J}$ is positive definite.*

Proof. To show $\mathcal{J}$ is positive definite, it suffices to verify that $\mathbf{c}' \mathcal{J} \mathbf{c} > 0$ holds for any nonzero constant vector $\mathbf{c} \in \mathbb{R}^n$. Recall that $\mathcal{J} = E(P_{0t}' S_2 P_{0t})$ in (A3.5) with

$$S_2 = \begin{pmatrix} \Sigma_0^{-1} & 0_{N \times N^2} \\ * & \frac{1}{2} \mathcal{P} \end{pmatrix}, \quad P_{0t} = \begin{pmatrix} \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} & 0_{N \times N(N+1)/2} \\ 0_{N^2 \times n_\alpha} & D \end{pmatrix},$$

$\mathcal{P} = \Sigma_0^{-1} \otimes \Sigma_0^{-1}$ and D being a duplication matrix. Note that Σ_0^{-1} is positive definite under Assumptions 3. Hence S_2 is positive definite, which implies that $\mathbf{c}' S_2 \mathbf{c} > 0$ holds for any nonzero constant vector $\mathbf{c} \in \mathbb{R}^n$. It follows that $\mathbf{c}' \mathcal{J} \mathbf{c} > 0$ holds for any $\mathbf{c}$ that satisfies $P_{0t} \mathbf{c} \neq 0_{(N+N^2) \times 1}$ a.s. Then we only need to show that $P_{0t} \mathbf{c} = 0_{(N+N^2) \times 1}$ holds a.s. if and only if $\mathbf{c} = 0_{n \times 1}$.

Suppose that $P_{0t} \mathbf{c} = 0_{(N+N^2) \times 1}$ a.s. with $\mathbf{c} = (\mathbf{c}'_1, \mathbf{c}'_2)'$, $\mathbf{c}_1 \in \mathbb{R}^{n_\alpha}$ and $\mathbf{c}_2 \in \mathbb{R}^{N(N+1)/2}$. It

follows that

$$\frac{\partial \boldsymbol{\varepsilon}_{0t}}{\partial \boldsymbol{\alpha}'} \mathbf{c}_1 = 0_{N \times 1} \text{ a.s.} \quad \text{and} \quad D \mathbf{c}_2 = 0_{N^2 \times 1}.$$

Recall that D is a duplication matrix. Since D has full column rank, $D \mathbf{c}_2 = 0_{N^2 \times 1}$ implies that $\mathbf{c}_2 = 0_{N(N+1)/2 \times 1}$. Moreover, by the proof of Lemma A13, $(\partial \boldsymbol{\varepsilon}_{0t} / \partial \boldsymbol{\alpha}') \mathbf{c}_1 = 0_{N \times 1}$ a.s. implies that $\mathbf{c}_1 = 0_{n_\alpha \times 1}$. Thus $P_{0t} \mathbf{c} = 0_{(N+N^2) \times 1}$ holds a.s. if and only if $\mathbf{c} = 0_{n \times 1}$. As a result, $\mathcal{J}$ is positive definite. $\square$

Lemma A8. *Suppose that $\{\mathbf{y}_t\}$ is generated by model (2.1) and Assumptions 1–4 hold. If $E(\|\boldsymbol{\varepsilon}_{0t}\|^4) < \infty$, $E(\|\mathbf{y}_t\|^2) < \infty$ and $\mathcal{I} = E[\partial l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta} \partial l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta}']$ is positive definite, then it holds that*

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial \tilde{l}_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \rightarrow_d N(0_{n \times 1}, \mathcal{I}) \quad \text{as } T \rightarrow \infty.$$

Proof. By (A3.3), Assumption 4 and Lemma A1(ii), it can be shown that

$$\begin{aligned} E \left[\left\| \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'} \right\| \right] &\leq E \left[\left\| \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\alpha}} \right\|^2 + \left\| \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\sigma}} \right\|^2 \right] \\ &\leq O(1) E \left[\left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \right\|^2 \|\boldsymbol{\varepsilon}_{0t}\|^2 + (1 + \|\boldsymbol{\varepsilon}_{0t}\|^2)^2 \right] \\ &\leq O(1) E \left[\xi_{\rho, t-1}^2 \|\boldsymbol{\varepsilon}_{0t}\|^2 + (1 + \|\boldsymbol{\varepsilon}_{0t}\|^2)^2 \right], \end{aligned}$$

where $\xi_{\rho t} = 1 + \sum_{h=0}^{\infty} \rho^h \|\mathbf{y}_{t-h}\|_2$. This together with $E(\|\boldsymbol{\varepsilon}_{0t}\|^4) < \infty$ and $E(\xi_{\rho t}^2) < \infty$ by $E(\|\mathbf{y}_t\|^2) < \infty$, Assumptions 2 and 4(ii), implies that $\mathcal{I}$ is finite.

Let $S_2 = \sum_{t=1}^T S_{2t}$ with $S_{2t} = \mathbf{c}' \partial l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta}$, where $\mathbf{c}$ is a constant vector with the same dimension as $\boldsymbol{\theta}$. Note that by Assumptions 1–3, $E(\boldsymbol{\varepsilon}_{0t} \mid \mathcal{F}_{t-1}) = \mathbf{0}$, $E(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}' \mid \mathcal{F}_{t-1}) = \Sigma_0$ (A3.1) and (A3.3), $\{S_{2t}\}$ is ergodic and stationary, S_{2t} is measurable with respect to $\mathcal{F}_t$ that is the σ -field generated by $\{\boldsymbol{\varepsilon}_{0s}, s \leq t\}$, and $E(S_{2t} \mid \mathcal{F}_{t-1}) = 0$. Moreover, since $\mathcal{I}$ is finite,

we can obtain that

$$\text{var}(S_{2t}) = \mathbf{c}' E[\partial l_t(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta} \partial l_t(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}'] \mathbf{c} = \mathbf{c}' \mathcal{I} \mathbf{c} < \infty.$$

Thus $\{S_{2t}, \mathcal{F}_t\}_t$ is an ergodic, stationary and square integrable martingale difference. By the martingale central limit theorem for a stationary and ergodic sequence of square integrable martingale increments (Billingsley, 1961), we have that

$$\frac{1}{\sqrt{T}} S_2 \rightarrow_d N(0, \mathbf{c}' \mathcal{I} \mathbf{c}) \text{ as } T \rightarrow \infty.$$

This together with the Cramér-Wold device, implies that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \rightarrow_d N(0_{n \times 1}, \mathcal{I}) \text{ as } T \rightarrow \infty. \quad (\text{A1.8})$$

Furthermore, by Lemma A4(ii), it holds that

$$\left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \left[\frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} - \frac{\partial \tilde{l}_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \right] \right\| \leq \frac{1}{\sqrt{T}} \sum_{t=1}^T O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|),$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$. Then by $0 < \rho < 1$ under Assumption 4(ii), $E(\xi_{\rho t}^2) < \infty$ and Markov's inequality, it can be shown that

$$\left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \left[\frac{\partial l_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} - \frac{\partial \tilde{l}_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \right] \right\| = O_p\left(\frac{1}{\sqrt{T}}\right). \quad (\text{A1.9})$$

Thus by (A1.8)–(A1.9), we can obtain that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial \tilde{l}_t(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}} \rightarrow_d N(0_{n \times 1}, \mathcal{I}) \text{ as } T \rightarrow \infty.$$

□

Lemma A9. Under Assumptions 2–4, it holds that

$$\begin{aligned} (i) & E \left(\sup_{\alpha \in \mathcal{A}} |\varepsilon'_t(\alpha) \varepsilon_t(\alpha)| \right) < \infty; \quad (ii) E \left(\sup_{\alpha \in \mathcal{A}} \left\| \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \varepsilon_t(\alpha) \right\| \right) < \infty; \\ (iii) & E \left(\sup_{\alpha \in \mathcal{A}} \left\| (\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha) / \partial \alpha)}{\partial \alpha'} + \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right\| \right) < \infty. \end{aligned}$$

Proof. We first show (i). Under Assumption 4, together with Lemma A1(i), it holds that

$$\sup_{\alpha \in \mathcal{A}} |\varepsilon'_t(\alpha) \varepsilon_t(\alpha)| \leq \sup_{\alpha \in \mathcal{A}} \|\varepsilon'_t(\alpha)\|^2 \leq O(1) (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|)^2.$$

Note that by Assumption 2 that $E(\|\mathbf{y}_t\|_2^2) < \infty$, we have $E(\xi_{\rho t}^2) < \infty$. These together with $E(\|\varepsilon_{0t}\|^2) < \infty$ by Assumption 3 imply that (i) holds.

For (ii), by Lemmas A1(i)–(ii), we have that

$$\begin{aligned} \sup_{\alpha \in \mathcal{A}} \left(\left\| \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \varepsilon_t(\alpha) \right\| \right) & < \sup_{\alpha \in \mathcal{A}} \left(\left\| \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \right\| \|\varepsilon_t(\alpha)\| \right) \\ & \leq O(1) \xi_{\rho, t-1} (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|) \leq O(1) (\xi_{\rho, t-1}^2 + \|\varepsilon_{0t}\|^2). \end{aligned}$$

Similarly, (ii) holds by $E(\xi_{\rho t}^2) < \infty$ and $E(\|\varepsilon_{0t}\|^2) < \infty$.

For (iii), under Assumption 4, by Lemma A1, it can be shown that

$$\begin{aligned} & \sup_{\alpha \in \mathcal{A}} \left\| (\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha) / \partial \alpha)}{\partial \alpha'} + \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right\| \\ & \leq O(1) \sup_{\alpha \in \mathcal{A}} \left(\|\varepsilon_t(\alpha)\| \left\| \frac{\partial (\text{vec}(\partial \varepsilon'_t(\alpha) / \partial \alpha))}{\partial \alpha'} \right\| + \left\| \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \right\| \left\| \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right\| \right) \\ & \leq O(1) [(\xi_{\rho, t-1} + \|\varepsilon_{0t}\|) \xi_{\rho, t-p-1} + \xi_{\rho, t-1}^2] \\ & \leq O(1) (\xi_{\rho, t-1}^2 + \|\varepsilon_{0t}\|^2). \end{aligned}$$

Then (iii) holds by $E(\xi_{\rho t}^2) < \infty$ and $E(\|\varepsilon_{0t}\|^2) < \infty$. □

Lemma A10. Under Assumptions 2–4, the following results hold:

$$(i) \sup_{\alpha \in \mathcal{A}} \left| \frac{1}{T} \sum_{t=1}^T \{ \varepsilon'_t(\alpha) \varepsilon_t(\alpha) - E[\varepsilon'_t(\alpha) \varepsilon_t(\alpha)] \} \right| = o_p(1);$$

$$\begin{aligned}
(ii) \quad & \sup_{\alpha \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \varepsilon_t(\alpha) - E \left[\frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \varepsilon_t(\alpha) \right] \right\} \right\| = o_p(1); \\
(iii) \quad & \sup_{\alpha \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \left[(\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha)/\partial \alpha)}{\partial \alpha'} + \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right] \right. \right. \\
& \quad \left. \left. - E \left[(\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha)/\partial \alpha)}{\partial \alpha'} + \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right] \right\} \right\| = o_p(1).
\end{aligned}$$

Proof. This lemma directly follows from Lemma A9 and Theorem 3.1 of Ling and McAleer (2003), so its proof is omitted here. $\square$

Lemma A11. *Under Assumption 4, the following results hold:*

$$\begin{aligned}
(i) \quad & \sup_{\alpha \in \mathcal{A}} |\varepsilon'_t(\alpha) \varepsilon_t(\alpha) - \tilde{\varepsilon}'_t(\alpha) \tilde{\varepsilon}_t(\alpha)| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|); \\
(ii) \quad & \sup_{\alpha \in \mathcal{A}} \left\| \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \varepsilon_t(\alpha) - \frac{\partial \tilde{\varepsilon}'_t(\alpha)}{\partial \alpha} \tilde{\varepsilon}_t(\alpha) \right\| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|); \\
(iii) \quad & \sup_{\alpha \in \mathcal{A}} \left\| \left[(\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha)/\partial \alpha)}{\partial \alpha'} \varepsilon_t(\alpha) + \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right] \right. \\
& \quad \left. - \left[(\tilde{\varepsilon}'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \tilde{\varepsilon}'_t(\alpha)/\partial \alpha)}{\partial \alpha'} + \frac{\partial \tilde{\varepsilon}_t(\alpha)}{\partial \alpha} \frac{\partial \tilde{\varepsilon}_t(\alpha)}{\partial \alpha'} \right] \right\| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|),
\end{aligned}$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$.

Proof. We first show (i). Recall that $\varepsilon_t(\alpha) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\alpha) \mathbf{y}_{t-h}$ and $\tilde{\varepsilon}_t(\alpha) = \mathbf{y}_t - \sum_{h=1}^{t-1} A_h(\alpha) \mathbf{y}_{t-h}$. By (A1.1) under Assumptions 4(ii)–(iii), it can be shown that

$$\sup_{\alpha \in \mathcal{A}} \|\varepsilon_t(\alpha) - \tilde{\varepsilon}_t(\alpha)\| \leq \sup_{\alpha \in \mathcal{A}} \sum_{h=t}^{\infty} \|A_h(\alpha)\| \|\mathbf{y}_{t-h}\|_2 \leq O(\rho^t) \xi_{\rho 0}. \quad (\text{A1.10})$$

This together with Lemma A1(i) implies that

$$\begin{aligned}
& \sup_{\alpha \in \mathcal{A}} |\varepsilon'_t(\alpha) \varepsilon_t(\alpha) - \tilde{\varepsilon}'_t(\alpha) \tilde{\varepsilon}_t(\alpha)| \\
& \leq \sup_{\alpha \in \mathcal{A}} |[\varepsilon_t(\alpha) - \tilde{\varepsilon}_t(\alpha)]' \varepsilon_t(\alpha)| + \sup_{\alpha \in \mathcal{A}} |\tilde{\varepsilon}'_t(\alpha) [\varepsilon_t(\alpha) - \tilde{\varepsilon}_t(\alpha)]| \leq O(\rho^t) \xi_{\rho 0} (\xi_{\rho, t-1} + \|\varepsilon_{0t}\|).
\end{aligned}$$

Hence (i) holds.

Similarly, by (A3.1)–(A3.2) and Assumptions 4(ii)–(iii), it can be shown that

$$\begin{aligned} \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} - \frac{\partial \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} \right\| &\leq O(\rho^{t-p-1})\xi_{\rho 0} \quad \text{and} \\ \sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{\partial(\text{vec}(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})/\partial \boldsymbol{\alpha}))}{\partial \boldsymbol{\alpha}'} - \frac{\partial(\text{vec}(\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})/\partial \boldsymbol{\alpha}))}{\partial \boldsymbol{\alpha}'} \right\| &\leq O(\rho^{t-p-2})\xi_{\rho 0}. \end{aligned} \quad (\text{A1.11})$$

These together with Lemma A1, imply that (ii)–(iii) hold. $\square$

Lemma A12. *Under Assumptions 1 and 3, $E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})]$ has a unique minimum at $\boldsymbol{\alpha}_0$.*

Proof. Recall that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha})\mathbf{y}_{t-h}$ and $\boldsymbol{\varepsilon}_{0t} = \mathbf{y}_t - \sum_{h=1}^{\infty} A_{0h}\mathbf{y}_{t-h}$. It holds that

$$E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] = E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})'(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})] + E[\boldsymbol{\varepsilon}'_{0t}\boldsymbol{\varepsilon}_{0t}] + 2E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})'\boldsymbol{\varepsilon}_{0t}].$$

Note that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t} = -\sum_{h=1}^{\infty} (A_h(\boldsymbol{\alpha}) - A_{0h})\mathbf{y}_{t-h} \in \mathcal{F}_{t-1}$ with $\mathcal{F}_t$ being the σ -field generated by $\{\boldsymbol{\varepsilon}_{0s}, s \leq t\}$, and $E(\boldsymbol{\varepsilon}_{0t} \mid \mathcal{F}_{t-1}) = 0$ under Assumption 3. Then using the law of iterated expectation, we have that

$$E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] = E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})'(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})] + E[\boldsymbol{\varepsilon}'_{0t}\boldsymbol{\varepsilon}_{0t}].$$

By Lemma A5, it can be easily shown that $E[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})'(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \boldsymbol{\varepsilon}_{0t})]$ has a unique minimum at $\boldsymbol{\alpha}_0$. It then follows that $E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})]$ has a unique minimum at $\boldsymbol{\alpha}_0$. $\square$

Lemma A13. *Under Assumptions 1 and 3, $\mathcal{I}_{LS}$ and $\mathcal{J}_{LS}$ are positive definite.*

Proof. We first show $\mathcal{J}_{LS}$ is positive definite, and it suffices to verify that $\mathbf{c}'\mathcal{J}_{LS}\mathbf{c} > 0$ holds for any nonzero constant vector $\mathbf{c} \in \mathbb{R}^{n_\alpha}$. Recall that $\mathcal{J}_{LS} = E[\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha} \partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}']$, hence $\mathbf{c}'\mathcal{J}_{LS}\mathbf{c} > 0$ holds for any $\mathbf{c}$ that satisfies $(\partial \boldsymbol{\varepsilon}_{0t}/\partial \boldsymbol{\alpha}')\mathbf{c} \neq 0_{N \times 1}$ a.s. Then we only need to show that $(\partial \boldsymbol{\varepsilon}_{0t}/\partial \boldsymbol{\alpha}')\mathbf{c} = 0_{N \times 1}$ holds a.s. if and only if $\mathbf{c} = 0_{n_\alpha \times 1}$. Suppose that $(\partial \boldsymbol{\varepsilon}_{0t}/\partial \boldsymbol{\alpha}')\mathbf{c} = 0_{N \times 1}$. Denote $\mathbf{y}_t = (y_{1t}, \dots, y_{Nt})'$. Note that (a) $\{y_{kt}, 1 \leq k \leq N\}$ are linearly independent, since if y_{kt} can be represented as a linear combination of $\{y_{k't}, k' \neq k\}$ then $\text{var}(\mathbf{y}_t \mid \mathcal{F}_{t-1}) = \Sigma_0$ is not full rank, which contradicts with the positive definiteness of Σ_0

under Assumption 3; (b) $\{\mathbf{y}_{t-j}, j \geq 0\}$ are linearly independent by the proof of Lemma A5. By (a)–(b) and (i) in the proof of Lemma A5, we can easily show that the following terms are linearly independent:

$$\left\{ \begin{aligned} &\sum_{m=1}^{\infty} m \lambda^{m-1} G_{p+i} y_{k,t-p-m}, \quad \sum_{m=1}^{\infty} m \gamma_j^{m-1} [\cos(m\varphi_j) G_{p+r+2j-1} + \sin(m\varphi_j) G_{p+r+2j}] y_{k,t-p-m}, \\ &\sum_{m=1}^{\infty} m \gamma_j^m [\sin(m\varphi_j) G_{p+r+2j-1} - \cos(m\varphi_j) G_{p+r+2j}] y_{k,t-p-m}, \quad y_{k,t-h}, \\ &\sum_{m=1}^{\infty} \lambda^m G_{p+i} y_{k,t-p-m}, \quad \text{and} \quad \sum_{m=1}^{\infty} \gamma_j^m [\cos(m\varphi_j) G_{p+r+2j-1} + \sin(m\varphi_j) G_{p+r+2j}] y_{k,t-p-m}, \\ &1 \leq h \leq p, 1 \leq i \leq r, 1 \leq k \leq N \end{aligned} \right\}.$$

This together with (A3.1) implies that $\mathbf{c} = 0_{n_\alpha \times 1}$. Thus $(\partial \boldsymbol{\varepsilon}_{0t} / \partial \boldsymbol{\alpha}') \mathbf{c} = 0_{N \times 1}$ holds a.s. if and only if $\mathbf{c} = 0_{n_\alpha \times 1}$. As a result, $\mathcal{I}_{LS}$ is positive definite.

Recall that $\mathcal{I}_{LS} = E[(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0) / \partial \boldsymbol{\alpha}) \Sigma_0 (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0) / \partial \boldsymbol{\alpha}')]$. Since Σ_0 is positive definite under Assumption 3, it can be similarly shown that $\mathcal{I}_{LS}$ is also positive definite. $\square$

Lemma A14. *Suppose that $\{\mathbf{y}_t\}$ is generated by model (2.1) and Assumptions 1–4 hold, then*

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \tilde{\boldsymbol{\varepsilon}}_{0t} \rightarrow_d N(0_{n_\alpha \times 1}, \mathcal{I}_{LS}) \quad \text{as } T \rightarrow \infty.$$

Proof. By Lemma A1(ii), it can be shown that

$$E \left(\left\| \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \Sigma_0 \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} \right\| \right) \leq E \left(\left\| \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \right\|^2 \|\Sigma_0\| \right) \leq O(1) E(\xi_{\rho,t-1}^2) \|\Sigma_0\|,$$

where $\xi_{\rho t} = 1 + \sum_{h=0}^{\infty} \rho^h \|\mathbf{y}_{t-h}\|_2$. This together with $\|\Sigma_0\| < \infty$ under Assumption 3, and $E(\xi_{\rho t}^2) < \infty$ by Assumptions 2 and 4(ii), implies that $\mathcal{I}_{LS}$ is finite.

Let $S_1 = \sum_{t=1}^T S_{1t}$ with $S_{1t} = \mathbf{c}' \partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0) / \partial \boldsymbol{\alpha} \boldsymbol{\varepsilon}_{0t}$, where $\mathbf{c}$ is a constant vector with the same dimension as $\boldsymbol{\alpha}$. Note that by Assumptions 1–3 and (A3.1), $\{S_{1t}\}$ is ergodic and stationary, S_{1t} is measurable with respect to $\mathcal{F}_t$ that is the σ -field generated by $\{\boldsymbol{\varepsilon}_{0s}, s \leq t\}$,

and $E(S_{1t} | \mathcal{F}_{t-1}) = 0$. Moreover, since $\mathcal{I}_{LS}$ is finite, we can obtain that

$$\text{var}(S_{1t}) = \mathbf{c}' E \left[\frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \Sigma_0 \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} \right] \mathbf{c} = \mathbf{c}' \mathcal{I}_{LS} \mathbf{c} < \infty.$$

Thus $\{S_{1t}, \mathcal{F}_t\}_t$ is an ergodic, stationary and square integrable martingale difference. By the martingale central limit theorem (CLT) for a stationary and ergodic sequence of square integrable martingale increments ([Billingsley, 1961](#)), we have that

$$\frac{1}{\sqrt{T}} S_1 \rightarrow_d N(0, \mathbf{c}' \mathcal{I}_{LS} \mathbf{c}) \text{ as } T \rightarrow \infty.$$

This together with the Cramér-Wold device, implies that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \boldsymbol{\varepsilon}_{0t} \rightarrow_d N(0_{n_\alpha \times 1}, \mathcal{I}_{LS}) \text{ as } T \rightarrow \infty. \quad (\text{A1.12})$$

Furthermore, by Lemma [A11\(ii\)](#), it holds that

$$\begin{aligned} \left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \left[\frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \boldsymbol{\varepsilon}_{0t} - \frac{\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \tilde{\boldsymbol{\varepsilon}}_{0t} \right] \right\| &\leq \frac{1}{\sqrt{T}} \sum_{t=1}^T \left\| \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \boldsymbol{\varepsilon}_{0t} - \frac{\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \tilde{\boldsymbol{\varepsilon}}_{0t} \right\| \\ &\leq \frac{1}{\sqrt{T}} \sum_{t=1}^T O(\rho^t) \xi_{\rho 0}(\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|). \end{aligned}$$

Then by $0 < \rho < 1$ under Assumption [4\(ii\)](#), $E(\xi_{\rho t}^2) < \infty$ and Markov's inequality, it can be shown that

$$\left\| \frac{1}{\sqrt{T}} \sum_{t=1}^T \left[\frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \boldsymbol{\varepsilon}_{0t} - \frac{\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \tilde{\boldsymbol{\varepsilon}}_{0t} \right] \right\| = O_p(T^{-1/2}). \quad (\text{A1.13})$$

Thus by [\(A1.12\)](#), [\(A1.13\)](#) and Slutsky's theorem, we can obtain that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \frac{\partial \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}} \tilde{\boldsymbol{\varepsilon}}_{0t} \rightarrow_d N(0_{n_\alpha \times 1}, \mathcal{I}_{LS}) \text{ as } T \rightarrow \infty.$$

□

A2 Technical Proofs

A2.1 Proof of Theorem 1

Proof. We first show the consistency of the QMLE $\hat{\boldsymbol{\theta}}$. By Theorem 4.1.1 in Amemiya (1985), it suffices to show the following results hold:

(A1) the space Θ is compact and $\boldsymbol{\theta}_0$ is an interior point in Θ ;

(A2) $\tilde{\mathcal{L}}(\boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta} \in \Theta$ and is a measurable function of $\{\mathbf{y}_t, t \in Z\}$ for all $\boldsymbol{\theta} \in \Theta$;

(A3) $\sup_{\boldsymbol{\theta} \in \Theta} |\tilde{\mathcal{L}}(\boldsymbol{\theta}) - E[l_t(\boldsymbol{\theta})]| \rightarrow_p 0$ as $T \rightarrow \infty$;

(A4) $E[l_t(\boldsymbol{\theta})]$ has a unique minimum at $\boldsymbol{\theta}_0$.

(A1) and (A2) hold under Assumptions 2 and 4(i), and (A4) holds by Lemma A6. Hence we only need to verify (A3). For (A3), recall that $\tilde{\mathcal{L}}(\boldsymbol{\theta}) = T^{-1} \sum_{t=1}^T \tilde{l}_t(\boldsymbol{\theta})$. By Lemma A4(i), we obtain that

$$\sup_{\boldsymbol{\theta} \in \Theta} |l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})| \leq O(\rho^t) \xi_{\rho 0} \xi_{\rho t}, \quad (\text{A2.1})$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$. Note that $0 < \rho < 1$ under Assumption 4(ii), and $E(\xi_{\rho t}^2) < \infty$ by $0 < \rho < 1$, $E(\|\mathbf{y}_t\|_2^2) < \infty$ and Assumption 2. Then using Markov's inequality, (A2.1) implies that

$$\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{1}{T} \sum_{t=1}^T [l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})] \right| \leq \frac{1}{T} \sum_{t=1}^T \sup_{\boldsymbol{\theta} \in \Theta} |l_t(\boldsymbol{\theta}) - \tilde{l}_t(\boldsymbol{\theta})| = O_p\left(\frac{1}{T}\right). \quad (\text{A2.2})$$

Moreover, by Lemma A3(i), it holds that

$$\sup_{\boldsymbol{\theta} \in \Theta} \left| \frac{1}{T} \sum_{t=1}^T \{l_t(\boldsymbol{\theta}) - E[l_t(\boldsymbol{\theta})]\} \right| = o_p(1) \text{ as } T \rightarrow \infty. \quad (\text{A2.3})$$

Thus by (A2.2) and (A2.3), we have that

$$\sup_{\boldsymbol{\theta} \in \Theta} |\tilde{\mathcal{L}}(\boldsymbol{\theta}) - E[l_t(\boldsymbol{\theta})]| \rightarrow_p 0 \text{ as } T \rightarrow \infty,$$

and then (A3) holds. As a result, Theorem 1(i) holds by (A1)–(A4).

Next we prove the asymptotic normality of $\widehat{\boldsymbol{\theta}}$. By Theorem 1(i), it holds that $\widehat{\boldsymbol{\theta}} \rightarrow_p \boldsymbol{\theta}_0$ as $T \rightarrow \infty$. To utilize Theorem 4.1.3 of Amemiya (1985), we then show that the following results hold:

(B1) $\partial^2 \widetilde{\mathcal{L}}(\boldsymbol{\theta})/(\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}')$ exists and is continuous on Θ ;

(B2) $\partial^2 \widetilde{\mathcal{L}}(\widehat{\boldsymbol{\theta}})/(\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}') \rightarrow_p \mathcal{J}$ as $T \rightarrow \infty$, and $\mathcal{J}$ is positive definite;

(B3) $T^{1/2} \partial \widetilde{\mathcal{L}}(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta} \rightarrow_d N(0_{n \times 1}, \mathcal{I})$ as $T \rightarrow \infty$.

(B1) holds under Assumption 2, and (B3) holds by Lemma A8. Hence we only need to show that (B2) holds. Similar to the proof of (A2.2), using Lemma A4(iii) and Markov's inequality, it can be shown that

$$\sup_{\boldsymbol{\theta} \in \Theta} \left\| \frac{1}{T} \sum_{t=1}^T \left(\frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} - \frac{\partial^2 \widetilde{l}_t(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right) \right\| = O_p \left(\frac{1}{T} \right).$$

This together with Lemma A3(iii) implies that

$$\frac{\partial^2 \widetilde{\mathcal{L}}(\widehat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \rightarrow_p E \left[\frac{\partial^2 l_t(\widehat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right] \text{ as } T \rightarrow \infty. \quad (\text{A2.4})$$

Moreover, since $\widehat{\boldsymbol{\theta}} \rightarrow_p \boldsymbol{\theta}_0$ as $T \rightarrow \infty$ and $\mathcal{J} = E(\partial^2 l_t(\boldsymbol{\theta}_0)/\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}')$ with $E(\partial^2 l_t(\boldsymbol{\theta})/\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}')$ being continuous on Θ , we can obtain that

$$E \left[\frac{\partial^2 l_t(\widehat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right] \rightarrow_p \mathcal{J} \text{ as } T \rightarrow \infty. \quad (\text{A2.5})$$

Then by (A2.4) and (A2.5), it holds that

$$\frac{\partial^2 \widetilde{\mathcal{L}}(\widehat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \rightarrow_p \mathcal{J} \text{ as } T \rightarrow \infty.$$

Furthermore, $\mathcal{J}$ is positive definite by Lemma A7. Thus (B2) holds. As a result, by the

consistency of $\widehat{\boldsymbol{\theta}}$, (B1)–(B3) and Theorem 4.1.3 of [Amemiya \(1985\)](#), it can be shown that

$$\sqrt{T}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(0_{n \times 1}, \mathcal{J}^{-1} \mathcal{I} \mathcal{J}^{-1}) \text{ as } T \rightarrow \infty.$$

This completes the proof of Theorem 1. □

A2.2 Proof of Proposition 1

Proof. We first show the consistency of the LSE $\widehat{\boldsymbol{\alpha}}_{LS}$. By Theorem 4.1.1 in [Amemiya \(1985\)](#), it suffices to show the following results hold:

- (a1) the space $\mathcal{A}$ is compact and $\boldsymbol{\alpha}_0$ is an interior point in $\mathcal{A}$;
- (a2) $\widetilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha})$ is continuous in $\boldsymbol{\alpha} \in \mathcal{A}$ and is a measurable function of $\{\mathbf{y}_t, t \in Z\}$ for all $\boldsymbol{\alpha} \in \mathcal{A}$;
- (a3) $\sup_{\boldsymbol{\alpha} \in \mathcal{A}} |\widetilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha}) - E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})]| \rightarrow_p 0$ as $T \rightarrow \infty$;
- (a4) $E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})]$ has a unique minimum at $\boldsymbol{\alpha}_0$.

(a1) and (a2) hold under Assumptions 2, 4(i) and Lemma A9(i), and (a4) holds by Lemma A12. Hence we only need to verify (a3). For (a3), recall that $\widetilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha}) = T^{-1} \sum_{t=1}^T \widetilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})\widetilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})$. By Lemma A11(i), we obtain that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} |\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \widetilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})\widetilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})| \leq O(\rho^t)\xi_{\rho 0}(\xi_{\rho, t-1} + \|\boldsymbol{\varepsilon}_{0t}\|), \quad (\text{A2.6})$$

where $\xi_{\rho t} = 1 + \sum_{j=0}^{\infty} \rho^j \|\mathbf{y}_{t-j}\|_2$. Note that $0 < \rho < 1$ under Assumption 4(ii), and $E(\xi_{\rho t}^2) < \infty$ by $0 < \rho < 1$, $E(\|\mathbf{y}_t\|_2^2) < \infty$ and Assumption 2. Then using Markov's inequality, (A2.6) implies that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left| \frac{1}{T} \sum_{t=1}^T [\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \widetilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})\widetilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})] \right| \leq \frac{1}{T} \sum_{t=1}^T \sup_{\boldsymbol{\alpha} \in \mathcal{A}} |\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - \widetilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})\widetilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha})| = O_p\left(\frac{1}{T}\right). \quad (\text{A2.7})$$

Moreover, by Lemma A10(i), it holds that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left| \frac{1}{T} \sum_{t=1}^T \{ \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) - E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] \} \right| = o_p(1) \text{ as } T \rightarrow \infty. \quad (\text{A2.8})$$

Thus by (A2.7) and (A2.8), we have that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left| \tilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha}) - E[\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] \right| \rightarrow_p 0 \text{ as } T \rightarrow \infty,$$

and then (a3) holds. As a result, the consistency of $\hat{\boldsymbol{\alpha}}_{LS}$ is established by (a1)–(a4).

Next we show the consistency of $\hat{\Sigma}_{LS}$. Recall that $\hat{\Sigma}_{LS} = T^{-1} \sum_{t=1}^T \tilde{\boldsymbol{\varepsilon}}_t(\hat{\boldsymbol{\alpha}}_{LS}) \tilde{\boldsymbol{\varepsilon}}'_t(\hat{\boldsymbol{\alpha}}_{LS})$. With similar arguments in the proof of Lemma A11 and (A2.7), we can show that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T [\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) - \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}) \tilde{\boldsymbol{\varepsilon}}'_t(\boldsymbol{\alpha})] \right\| = O_p \left(\frac{1}{T} \right).$$

This together with $\hat{\boldsymbol{\alpha}}_{LS} \rightarrow_p \boldsymbol{\alpha}_0$ as $T \rightarrow \infty$ and $\boldsymbol{\alpha}_0$ is an interior point in $\mathcal{A}$ under Assumption 4(i), implies that

$$\left\| \hat{\Sigma}_{LS} - \frac{1}{T} \sum_{t=1}^T \boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}_{LS}) \boldsymbol{\varepsilon}'_t(\hat{\boldsymbol{\alpha}}_{LS}) \right\| = O_p \left(\frac{1}{T} \right).$$

And it follows that

$$\hat{\Sigma}_{LS} - \frac{1}{T} \sum_{t=1}^T \boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}_{LS}) \boldsymbol{\varepsilon}'_t(\hat{\boldsymbol{\alpha}}_{LS}) = O_p \left(\frac{1}{T} \right). \quad (\text{A2.9})$$

Moreover, by Taylor's expansion and $\boldsymbol{\varepsilon}_{0t} = \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)$, we have that

$$\begin{aligned} & \frac{1}{T} \sum_{t=1}^T \text{vec} [\boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}_{LS}) \boldsymbol{\varepsilon}'_t(\hat{\boldsymbol{\alpha}}_{LS})] \\ &= \frac{1}{T} \sum_{t=1}^T \left\{ \text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t}) + \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} \right] (\hat{\boldsymbol{\alpha}}_{LS} - \boldsymbol{\alpha}_0) \right\} \end{aligned} \quad (\text{A2.10})$$

where $\boldsymbol{\alpha}^*$ lies between $\hat{\boldsymbol{\alpha}}_{LS}$ and $\boldsymbol{\alpha}_0$. With analogous arguments in the proof of Lemma A10,

it can be shown that

$$\sup_{\boldsymbol{\alpha} \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} \right] - E \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} \right] \right\} \right\| = o_p(1), \quad (\text{A2.11})$$

as $T \rightarrow \infty$. Note that by (A3.1), $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$ and $E[\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)] = 0$ under Assumption 3 and using the law of iterated expectation, we have that

$$E \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)}{\partial \boldsymbol{\alpha}'} \right] = 0. \quad (\text{A2.12})$$

Then by (A2.11)–(A2.12), $\widehat{\boldsymbol{\alpha}}_{LS} \rightarrow_p \boldsymbol{\alpha}_0$ as $T \rightarrow \infty$ and $\boldsymbol{\alpha}_0$ is an interior point in $\mathcal{A}$, it can be obtained that

$$\frac{1}{T} \sum_{t=1}^T \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} \right] = o_p(1) \text{ as } T \rightarrow \infty, \quad (\text{A2.13})$$

and thus (A2.10) implies that

$$\frac{1}{T} \sum_{t=1}^T \text{vec}[\boldsymbol{\varepsilon}_t(\widehat{\boldsymbol{\alpha}}_{LS}) \boldsymbol{\varepsilon}_t'(\widehat{\boldsymbol{\alpha}}_{LS})] = \frac{1}{T} \sum_{t=1}^T \text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}') + o_p(1) \text{ as } T \rightarrow \infty. \quad (\text{A2.14})$$

Furthermore, using the law of large number under Assumption 3, it holds that

$$\frac{1}{T} \sum_{t=1}^T \boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}' - \Sigma_0 = \frac{1}{T} \sum_{t=1}^T \boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}' - \frac{1}{T} \sum_{t=1}^T E(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}_{0t}') = o_p(1) \text{ as } T \rightarrow \infty. \quad (\text{A2.15})$$

As a result, (A2.9), (A2.14) and (A2.15) imply that $\widehat{\Sigma}_{LS} \rightarrow_p \Sigma_0$ as $T \rightarrow \infty$. Then Proposition 1(i) is established.

We then prove the asymptotic normality of $\widehat{\boldsymbol{\alpha}}_{LS}$. By Proposition 1(i), it holds that $\widehat{\boldsymbol{\alpha}}_{LS} \rightarrow_p \boldsymbol{\alpha}_0$ as $T \rightarrow \infty$. To utilize Theorem 4.1.3 of Amemiya (1985), we then show that the following results hold:

(b1) $\partial^2 \widetilde{\mathcal{L}}_{LS}(\boldsymbol{\alpha}) / (\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}')$ exists and is continuous on Θ ;

(b2) $\partial^2 \tilde{\mathcal{L}}_{LS}(\hat{\alpha}_{LS})/(\partial \alpha \partial \alpha') \rightarrow_p \mathcal{J}_{LS}$ as $T \rightarrow \infty$, and $\mathcal{J}_{LS}$ is positive definite;

(b3) $T^{1/2} \partial \tilde{\mathcal{L}}_{LS}(\alpha_0)/\partial \alpha \rightarrow_d N(0_{n_\alpha \times 1}, \mathcal{I}_{LS})$ as $T \rightarrow \infty$.

(b1) holds under Assumption 2, and (b3) holds by Lemma A14. Hence we only need to show that (b2) holds. Similar to the proof of (A2.7), using Lemma A11(iii) and Markov's inequality, it can be shown that

$$\sup_{\alpha \in \mathcal{A}} \left\| \frac{1}{T} \sum_{t=1}^T \left\{ \left[(\varepsilon'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \varepsilon'_t(\alpha)/\partial \alpha)}{\partial \alpha'} \varepsilon_t(\alpha) + \frac{\partial \varepsilon'_t(\alpha)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha)}{\partial \alpha'} \right] - \left[(\tilde{\varepsilon}'_t(\alpha) \otimes I_{n_\alpha}) \frac{\partial \text{vec}(\partial \tilde{\varepsilon}'_t(\alpha)/\partial \alpha)}{\partial \alpha'} + \frac{\partial \tilde{\varepsilon}'_t(\alpha)}{\partial \alpha} \frac{\partial \tilde{\varepsilon}_t(\alpha)}{\partial \alpha'} \right] \right\} \right\| = O_p\left(\frac{1}{T}\right).$$

This together with Lemma A10(iii) implies that

$$\frac{\partial^2 \tilde{\mathcal{L}}_{LS}(\hat{\alpha}_{LS})}{\partial \alpha \partial \alpha'} \rightarrow_p E \left[\frac{\partial^2 \mathcal{L}_{LS}(\hat{\alpha}_{LS})}{\partial \alpha \partial \alpha'} \right] \text{ as } T \rightarrow \infty. \quad (\text{A2.16})$$

Moreover, note that

$$E \left[\frac{\partial^2 \mathcal{L}_{LS}(\alpha_0)}{\partial \alpha \partial \alpha'} \right] = E \left[\frac{\partial \varepsilon'_t(\alpha_0)}{\partial \alpha} \frac{\partial \varepsilon_t(\alpha_0)}{\partial \alpha'} \right] = \mathcal{J}_{LS}.$$

Since $\hat{\alpha}_{LS} \rightarrow_p \alpha_0$ as $T \rightarrow \infty$ and $E[\partial^2 \mathcal{L}_{LS}(\alpha)/\partial \alpha \partial \alpha']$ is continuous on $\mathcal{A}$, we have that

$$E \left[\frac{\partial^2 \tilde{\mathcal{L}}_{LS}(\hat{\alpha}_{LS})}{\partial \alpha \partial \alpha'} \right] \rightarrow_p \mathcal{J}_{LS} \text{ as } T \rightarrow \infty. \quad (\text{A2.17})$$

Then by (A2.16) and (A2.17), it holds that

$$\frac{\partial^2 \tilde{\mathcal{L}}_{LS}(\hat{\alpha}_{LS})}{\partial \alpha \partial \alpha'} \rightarrow_p \mathcal{J}_{LS} \text{ as } T \rightarrow \infty.$$

Furthermore, $\mathcal{J}_{LS}$ is positive definite by Lemma A13. Thus (b2) holds. As a result, by the consistency of $\hat{\alpha}_{LS}$, (b1)–(b3) and Theorem 4.1.3 of Amemiya (1985), it can be shown that

$$\sqrt{T}(\hat{\alpha}_{LS} - \alpha_0) \rightarrow_d N(0_{n_\alpha \times 1}, \mathcal{J}_{LS}^{-1} \mathcal{I}_{LS} \mathcal{J}_{LS}^{-1}) \text{ as } T \rightarrow \infty,$$

and then Proposition 1(ii) holds.

Lastly we prove Proposition 1(iii), that is the asymptotic normality of $\text{vec}(\widehat{\Sigma}_{LS})$. By (A2.9)–(A2.10), as $T \rightarrow \infty$, it holds that

$$\begin{aligned} & \sqrt{T} \text{vec}(\widehat{\Sigma}_{LS}) \\ &= \frac{1}{\sqrt{T}} \sum_{t=1}^T \left\{ \text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t}) + \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} \right] (\widehat{\boldsymbol{\alpha}}_{LS} - \boldsymbol{\alpha}_0) \right\} + o_p(1), \end{aligned} \quad (\text{A2.18})$$

where $\boldsymbol{\alpha}^*$ lies between $\widehat{\boldsymbol{\alpha}}_{LS}$ and $\boldsymbol{\alpha}_0$. Using Lindeberg–Lévy central limit theorem under Assumption 3, $E(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t} \mid \mathcal{F}_{t-1}) = \Sigma_0$ and $E(\|\boldsymbol{\varepsilon}_{0t}\|^4) < \infty$, then by the martingale CLT for a stationary and ergodic sequence of square integrable martingale increments (Billingsley, 1961) we obtain that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T \text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t} - \Sigma_0) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K}) \text{ as } T \rightarrow \infty,$$

where $\mathcal{K} = \text{var}[\text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t})]$. Moreover, by Proposition 1(ii) and (A2.13), it holds that

$$\frac{1}{T} \sum_{t=1}^T \left[(\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*) \otimes I_N) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} + (I_N \otimes \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}'} \right] \sqrt{T} (\widehat{\boldsymbol{\alpha}}_{LS} - \boldsymbol{\alpha}_0) = o_p(1) \quad (\text{A2.19})$$

as $T \rightarrow \infty$. As a result, using Slutsky's theorem, (A2.18)–(A2.19) imply that

$$\sqrt{T} \left(\text{vec}(\widehat{\Sigma}_{LS}) - \text{vec}(\Sigma_0) \right) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K}) \text{ as } T \rightarrow \infty.$$

This completes the proof of Proposition 1. □

A2.3 Proof of Proposition 2

Proof. We first establish the following results:

$$(a) \quad \sqrt{T}(\widehat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_0) \rightarrow_d N(0_{n_\alpha \times 1}, \Xi) \text{ as } T \rightarrow \infty;$$

(b) $\sqrt{T}(\text{vec}(\widehat{\Sigma}) - \text{vec}(\Sigma_0)) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K})$ as $T \rightarrow \infty$,

where $\boldsymbol{\theta}_0 = (\boldsymbol{\alpha}'_0, \boldsymbol{\sigma}'_0)'$ with $\boldsymbol{\sigma}_0 = \text{vech}(\Sigma_0)$, $\widehat{\boldsymbol{\theta}} = (\widehat{\boldsymbol{\alpha}}', \widehat{\boldsymbol{\sigma}}')'$ with $\widehat{\boldsymbol{\sigma}} = \text{vech}(\widehat{\Sigma})$, $\Xi = \{E[(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) \Sigma_0^{-1}(\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')] \}^{-1}$ and $\mathcal{K} = \text{var}[\text{vec}(\boldsymbol{\varepsilon}_{0t} \boldsymbol{\varepsilon}'_{0t})]$. For (a), by Theorem 1, it holds that

$$\sqrt{T}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow_d N(0_{n \times 1}, \mathcal{J}^{-1} \mathcal{I} \mathcal{J}^{-1}), \quad (\text{A2.20})$$

where $\mathcal{I} = E(P'_{0t} S_1 P_{0t})$ and $\mathcal{J} = E(P'_{0t} S_2 P_{0t})$ with S_1, S_2 and P_{0t} block structured by (A3.5). By matrix multiplication, it can be directly obtained that the asymptotic covariance matrix of $\sqrt{T}(\widehat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_0)$ is Ξ . This together with (A2.20) implies that (a) holds. For (b), note that $\widehat{\boldsymbol{\theta}} = (\widehat{\boldsymbol{\alpha}}', \widehat{\boldsymbol{\sigma}}')'$ satisfies that

$$0 = \sum_{t=1}^T \frac{\partial \widetilde{l}_t(\widehat{\boldsymbol{\theta}})}{\partial \boldsymbol{\sigma}} = \frac{1}{2} D' \sum_{t=1}^T \text{vec} \left(\Sigma^{-1}(\widehat{\boldsymbol{\sigma}}) - \Sigma^{-1}(\widehat{\boldsymbol{\sigma}}) \widetilde{\boldsymbol{\varepsilon}}_t(\widehat{\boldsymbol{\alpha}}) \widetilde{\boldsymbol{\varepsilon}}'_t(\widehat{\boldsymbol{\alpha}}) \Sigma^{-1}(\widehat{\boldsymbol{\sigma}}) \right),$$

where D is a duplication matrix and $\widetilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{t-1} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$. This together with $\widehat{\boldsymbol{\sigma}} = \text{vech}(\widehat{\Sigma})$ and $\widehat{\Sigma} = \Sigma(\widehat{\boldsymbol{\sigma}})$ implies that

$$\widehat{\Sigma} = \frac{1}{T} \sum_{t=1}^T \widetilde{\boldsymbol{\varepsilon}}_t(\widehat{\boldsymbol{\alpha}}) \widetilde{\boldsymbol{\varepsilon}}'_t(\widehat{\boldsymbol{\alpha}}).$$

Moreover, by Proposition 1, it holds that

$$\widehat{\Sigma}_{LS} = \frac{1}{T} \sum_{t=1}^T \widetilde{\boldsymbol{\varepsilon}}_t(\widehat{\boldsymbol{\alpha}}_{LS}) \widetilde{\boldsymbol{\varepsilon}}'_t(\widehat{\boldsymbol{\alpha}}_{LS}) \quad \text{and} \quad \sqrt{T}(\text{vec}(\widehat{\Sigma}_{LS}) - \text{vec}(\Sigma_0)) \rightarrow_d N(0_{N^2 \times 1}, \mathcal{K}) \quad (\text{A2.21})$$

where the asymptotic distribution depends on the consistency of $\widehat{\boldsymbol{\alpha}}_{LS}$. Thus $\widehat{\Sigma}$ has the same asymptotic distribution as $\widehat{\Sigma}_{LS}$, and then (b) holds.

Next we show (i). Recall that $\Xi_1 = \mathcal{J}_{LS}^{-1} \mathcal{I}_{LS} \mathcal{J}_{LS}^{-1}$ with $\mathcal{I}_{LS} = E[(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) \Sigma_0 (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')]$ and $\mathcal{J}_{LS} = E[(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')]$, and $\Xi = \left\{ E[(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) \Sigma_0^{-1} (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')] \right\}^{-1}$.

Denote the random matrix

$$\mathcal{R}_t = \begin{bmatrix} A_t & -B_t \\ -B_t & C_t \end{bmatrix},$$

where $A_t = (\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) \Sigma_0^{-1} (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')$, $B_t = (\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')$, and $C_t = (\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}) \Sigma_0 (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}')$. Then it holds that $\Xi_1 - \Xi = (EB_t)^{-1} (EC_t) (EB_t)^{-1} - (EA_t)^{-1}$, and $(EB_t)(\Xi_1 - \Xi)(EB_t) = EC_t - (EB_t)(EA_t)^{-1}(EB_t)$ is the Schur complement of the block EA_t of the block matrix ER_t . Note that by Lemma A13 and its proof, matrices EA_t , EB_t and EC_t are positive definite. Thus using the property of the Schur complement of a block matrix, to show that $\Xi_1 - \Xi$ is positive semi-definite, it suffices to show that ER_t is positive semi-definite. Let $\mathbf{c} = (\mathbf{c}'_1, \mathbf{c}'_2)'$ be an arbitrary constant vector with $\mathbf{c}_1, \mathbf{c}_2 \in \mathbb{R}^{n_\alpha}$. We have that $\mathbf{c}' \mathcal{R}_t \mathbf{c} = \mathbf{c}'_1 A_t \mathbf{c}_1 - 2\mathbf{c}'_1 B_t \mathbf{c}_2 + \mathbf{c}'_2 C_t \mathbf{c}_2$. Denote $\tilde{\mathbf{c}}_1 = \Sigma_0^{-1/2} (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}') \mathbf{c}_1$ and $\tilde{\mathbf{c}}_2 = \Sigma_0^{1/2} (\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0)/\partial \boldsymbol{\alpha}') \mathbf{c}_2$, it then follows that $\mathbf{c}' \mathcal{R}_t \mathbf{c} = \tilde{\mathbf{c}}'_1 \tilde{\mathbf{c}}_1 - 2\tilde{\mathbf{c}}'_1 \tilde{\mathbf{c}}_2 + \tilde{\mathbf{c}}'_2 \tilde{\mathbf{c}}_2 = (\tilde{\mathbf{c}}_1 - \tilde{\mathbf{c}}_2)' (\tilde{\mathbf{c}}_1 - \tilde{\mathbf{c}}_2) \geq 0$ a.s. Thus $\mathbf{c}' ER_t \mathbf{c} \geq 0$, which implies that ER_t is positive semi-definite. As a result, $\Xi_1 - \Xi$ is positive semi-definite. Moreover, it is obvious that $\Xi_1 = \Xi$ holds if and only if $\Sigma_0 = \sigma^2 I_N$ for some $\sigma^2 > 0$.

Note that (ii) can be verified directly by (b) and (A2.21). We accomplish the proof of this proposition. $\square$

A2.4 Proofs of Remark 1 and Theorem 2

Proof of Remark 1. By (i) and (ii) in the proof of Lemma A5, Assumption 1 ensures the orders r and s non-degenerate.

Next we show that the condition $G_{0p} \neq \sum_{i=1}^r G_{0,p+i} + \sum_{j=1}^s G_{0,p+r+2j-1}$ guarantees the order p non-decreasing. Note that the order p will reduce to $p-1$ if and only if there exist $\{\lambda_{*i}\}_{i=1}^r$, $\{\gamma_{*j}\}_{j=1}^s$, $\{\varphi_{*j}\}_{j=1}^s$ and $\{G_{*k}\}_{k=p}^{p+r+2s}$ such that the following equations holds:

$$\begin{aligned}
G_{0p} &= \sum_{i=1}^r \lambda_{*i} G_{*,p+i-1} + \sum_{j=1}^s \gamma_{*j} [\cos(\varphi_{*j}) G_{*,p+r+2j-2} + \sin(\varphi_{*j}) G_{*,p+r+2j-1}], \quad (\text{A2.22}) \\
&\sum_{i=1}^r \lambda_{0i}^{h-p} G_{0,p+i} + \sum_{j=1}^s \gamma_{0j}^{h-p} [\cos((h-p)\varphi_{0j}) G_{0,p+r+2j-1} + \sin((h-p)\varphi_{0j}) G_{0,p+r+2j}] \\
&= \sum_{i=1}^r \lambda_{*i}^{h-p+1} G_{*,p+i-1} + \sum_{j=1}^s \gamma_{*j}^{h-p+1} [\cos((h-p+1)\varphi_{*j}) G_{*,p+r+2j-2} \\
&\quad + \sin((h-p+1)\varphi_{*j}) G_{*,p+r+2j-1}]
\end{aligned}$$

$$\text{for all } h \geq p + 1. \quad (\text{A2.23})$$

By (i) and (ii) in the proof of Lemma A5, (A2.23) implies that $\lambda_{*i} = \lambda_{0i}$, $\gamma_{*j} = \gamma_{0j}$, $\varphi_{*j} = \varphi_{0j}$, $\lambda_{*i}G_{*,p+i-1} = G_{0,p+i}$, and

$$\begin{aligned} & \gamma_{*j} [\cos((h-p+1)\varphi_{*j})G_{*,p+r+2j-2} + \sin((h-p+1)\varphi_{*j})G_{*,p+r+2j-1}] \\ &= \cos((h-p)\varphi_{0j})G_{0,p+r+2j-1} + \sin((h-p)\varphi_{0j})G_{0,p+r+2j}, \end{aligned} \quad (\text{A2.24})$$

hold for all $1 \leq i \leq r$, $1 \leq j \leq s$ and $h \geq p + 1$. Furthermore, using the fact $a \cos((m+1)\varphi) + b \sin((m+1)\varphi) = (a \cos(\varphi) + b \sin(\varphi)) \cos(m\varphi) + (b \cos(\varphi) - a \sin(\varphi)) \sin(m\varphi)$, (A2.24) implies that

$$\begin{aligned} & \gamma_{*j} (\cos(\varphi_{*j})G_{*,p+r+2j-2} + \sin(\varphi_{*j})G_{*,p+r+2j-1}) \cos((h-p)\varphi_{*j}) \\ &+ \gamma_{*j} (\cos(\varphi_{*j})G_{*,p+r+2j-1} - \sin(\varphi_{*j})G_{*,p+r+2j-2}) \sin((h-p)\varphi_{*j}) \\ &= G_{0,p+r+2j-1} \cos((h-p)\varphi_{0j}) + G_{0,p+r+2j} \sin((h-p)\varphi_{0j}), \end{aligned}$$

for all $1 \leq j \leq s$ and $h \geq p + 1$, and then by $\gamma_{*j} = \gamma_{0j}$ and $\varphi_{*j} = \varphi_{0j}$, we have that

$$\begin{aligned} & \cos(\varphi_{*j})G_{*,p+r+2j-2} + \sin(\varphi_{*j})G_{*,p+r+2j-1} = \gamma_{*j}^{-1}G_{0,p+r+2j-1} \quad \text{and} \quad (\text{A2.25}) \\ & \cos(\varphi_{*j})G_{*,p+r+2j-1} - \sin(\varphi_{*j})G_{*,p+r+2j-2} = \gamma_{*j}^{-1}G_{0,p+r+2j}, \end{aligned}$$

for all $1 \leq j \leq s$ and $h \geq p + 1$. Above all, by $\lambda_{*i}G_{*,p+i-1} = G_{0,p+i}$, (A2.25) and (A2.22), it can be obtained that

$$G_{0p} = \sum_{i=1}^r G_{0,p+i} + \sum_{j=1}^s G_{0,p+r+2j-1}.$$

As a result, if $G_{0p} \neq \sum_{i=1}^r G_{0,p+i} + \sum_{j=1}^s G_{0,p+r+2j-1}$, then (A2.22)–(A2.23) do not hold, which implies that the order p is irreducible. $\square$

Proof of Theorem 2. Recall that $\theta = (\alpha', \sigma')'$. For $\iota = (p, r, s)$, $\iota^* = (p^*, r^*, s^*) \in \mathcal{M} =$

$\{\iota \mid 0 \leq p \leq p_{\max}, 0 \leq r \leq r_{\max}, 0 \leq s \leq s_{\max}\}$, denote Θ_ι (or Θ_{ι^*}) as the parameter space of $\boldsymbol{\theta}$ with the order set to ι (or ι^*), and $\mathcal{A}_\iota$ (or $\mathcal{A}_{\iota^*}$) as the parameter space of $\boldsymbol{\alpha}$ with the order set to ι (or ι^*). Let $\hat{\boldsymbol{\alpha}}_\iota$ (or $\hat{\boldsymbol{\alpha}}_{\iota^*}$) be the LSE $\hat{\boldsymbol{\alpha}}_1$ or QMLE $\hat{\boldsymbol{\alpha}}_2$ with the order set to ι (or ι^*). To prove the selection consistency of the proposed BIC in (3.6), it suffices to show that the following result holds for any $\iota \neq \iota^*$:

$$\lim_{T \rightarrow \infty} P(\text{BIC}(\iota) - \text{BIC}(\iota^*) > 0) = 1. \quad (\text{A2.26})$$

By (3.6), it holds that

$$\text{BIC}(\iota) - \text{BIC}(\iota^*) = T \left(\ln |\hat{\Sigma}(\hat{\boldsymbol{\alpha}}_\iota)| - \ln |\hat{\Sigma}(\hat{\boldsymbol{\alpha}}_{\iota^*})| \right) + (n(\iota) - n(\iota^*)) \ln T, \quad (\text{A2.27})$$

where $\hat{\Sigma}(\boldsymbol{\alpha}) = T^{-1} \sum_{t=1}^T \tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}) \tilde{\boldsymbol{\varepsilon}}_t'(\boldsymbol{\alpha})$ with $\tilde{\boldsymbol{\varepsilon}}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{t-1} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$. By Lemma A6, we have that

$$\boldsymbol{\theta}_0 = (\boldsymbol{\alpha}'_0, \boldsymbol{\sigma}'_0)' = \arg \min_{\boldsymbol{\theta} \in \Theta_{\iota^*}} \left\{ \ln |\Sigma(\boldsymbol{\sigma})| + E[\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] \right\},$$

with $\Sigma_0 = \Sigma(\boldsymbol{\sigma}_0)$. In addition, let $\mathring{\Sigma} = \Sigma(\mathring{\boldsymbol{\sigma}})$ and

$$\mathring{\boldsymbol{\theta}} = (\mathring{\boldsymbol{\alpha}}', \mathring{\boldsymbol{\sigma}}')' = \arg \min_{\boldsymbol{\theta} \in \Theta_\iota} \left\{ \ln |\Sigma(\boldsymbol{\sigma})| + E[\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})' \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})] \right\}.$$

To verify (A2.26), we next consider two cases.

Case I (overfitting): $p \geq p^*, r \geq r^*, s \geq s^*$, and at least one inequality holds. Note that $n(\iota) - n(\iota^*) > 0$ in Case I, which implies that $(n(\iota) - n(\iota^*)) \ln T \rightarrow \infty$ as $T \rightarrow \infty$. Thus by (A2.27), to establish that (A2.26) holds for Case I, it suffices to show that

$$T(\ln |\hat{\Sigma}(\hat{\boldsymbol{\alpha}}_\iota)| - \ln |\hat{\Sigma}(\hat{\boldsymbol{\alpha}}_{\iota^*})|) = O_p(1) \text{ as } T \rightarrow \infty.$$

Rewrite $\ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - \ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_{\iota^*})|$ as follows,

$$\begin{aligned} \ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - \ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_{\iota^*})| &= \left(\ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - \ln |\Sigma(\dot{\boldsymbol{\alpha}})| \right) + \left(\ln |\Sigma(\dot{\boldsymbol{\alpha}})| - \ln |\Sigma(\boldsymbol{\alpha}_0)| \right) \\ &\quad + \left(\ln |\Sigma(\boldsymbol{\alpha}_0)| - \ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_{\iota^*})| \right), \end{aligned} \quad (\text{A2.28})$$

where $\Sigma(\boldsymbol{\alpha}) = T^{-1} \sum_{t=1}^T \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}_t'(\boldsymbol{\alpha})$ with $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\boldsymbol{\alpha}) \mathbf{y}_{t-h}$. Notice that the model with order $\iota = (p, r, s)$ in Case I corresponds to a bigger model. It holds that $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}_0) = \boldsymbol{\varepsilon}_t(\dot{\boldsymbol{\alpha}})$ for all t , which implies that $\Sigma(\boldsymbol{\alpha}_0) = \Sigma(\dot{\boldsymbol{\alpha}})$ and then the second parenthesis in (A2.28) is zero. Below we will show that the following result holds when $p \geq p^*, r \geq r^*, s \geq s^*$:

$$\ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - \ln |\Sigma(\dot{\boldsymbol{\alpha}})| = O_p(T^{-1}), \quad (\text{A2.29})$$

which implies that the first and last parenthesis are $O_p(T^{-1})$. Since it holds that

$$\ln |\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - \ln |\Sigma(\dot{\boldsymbol{\alpha}})| = \ln \left(1 + \frac{|\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - |\Sigma(\dot{\boldsymbol{\alpha}})|}{|\Sigma(\dot{\boldsymbol{\alpha}})|} \right), \quad (\text{A2.30})$$

we only need to verify that (a) $|\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota)| - |\Sigma(\dot{\boldsymbol{\alpha}})| = O_p(T^{-1})$; and (b) $|\Sigma(\dot{\boldsymbol{\alpha}})| = O_p(1)$. In addition, as Case I corresponds to a bigger model, we set $G_{0,p+i} = 0, G_{0,p+r+2j-1} = 0, G_{0,p+r+2j} = 0$ for $r^* < i \leq r$ and $s^* < j \leq s$ in this case.

We first show (a). It suffices to show that $\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota) - \Sigma(\dot{\boldsymbol{\alpha}}) = O_p(T^{-1})$ holds. By Lemma A11, we have

$$\widehat{\Sigma}(\widehat{\boldsymbol{\alpha}}_\iota) - \Sigma(\widehat{\boldsymbol{\alpha}}_\iota) = O_p(T^{-1}). \quad (\text{A2.31})$$

Then we are left to establish that

$$\Sigma(\widehat{\boldsymbol{\alpha}}_\iota) - \Sigma(\dot{\boldsymbol{\alpha}}) = O_p(T^{-1}). \quad (\text{A2.32})$$

Denote $\widehat{\Delta}_h = A_h(\widehat{\boldsymbol{\alpha}}_\iota) - A_h(\dot{\boldsymbol{\alpha}})$. It can be shown that

$$T [\Sigma(\widehat{\boldsymbol{\alpha}}_\iota) - \Sigma(\dot{\boldsymbol{\alpha}})] = \sum_{t=1}^T [\boldsymbol{\varepsilon}_t(\widehat{\boldsymbol{\alpha}}_\iota) \boldsymbol{\varepsilon}_t'(\widehat{\boldsymbol{\alpha}}_\iota) - \boldsymbol{\varepsilon}_t(\dot{\boldsymbol{\alpha}}) \boldsymbol{\varepsilon}_t'(\dot{\boldsymbol{\alpha}})]$$

$$\begin{aligned}
&= \sum_{t=1}^T \left[\left(\boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}) - \sum_{h=1}^{\infty} \hat{\Delta}_h \mathbf{y}_{t-h} \right) \left(\boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}) - \sum_{h=1}^{\infty} \hat{\Delta}_h \mathbf{y}_{t-h} \right)' - \boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}) \boldsymbol{\varepsilon}_t'(\hat{\boldsymbol{\alpha}}) \right] \\
&= \sum_{t=1}^T \left[- \sum_{h=1}^{\infty} \boldsymbol{\varepsilon}_t(\hat{\boldsymbol{\alpha}}) \mathbf{y}_{t-h}' \hat{\Delta}_h' - \sum_{h=1}^{\infty} \hat{\Delta}_h \mathbf{y}_{t-h} \boldsymbol{\varepsilon}_t'(\hat{\boldsymbol{\alpha}}) + \sum_{h_1=1}^{\infty} \sum_{h_2=1}^{\infty} \hat{\Delta}_{h_1} \mathbf{y}_{t-h_1} \mathbf{y}_{t-h_2}' \hat{\Delta}_{h_2}' \right] \\
&= - \left(\sum_{h=1}^{\infty} \sqrt{T} \hat{\Delta}_h \frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{y}_{t-h} \boldsymbol{\varepsilon}_t'(\hat{\boldsymbol{\alpha}}) \right)' - \sum_{h=1}^{\infty} \sqrt{T} \hat{\Delta}_h \frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{y}_{t-h} \boldsymbol{\varepsilon}_t'(\hat{\boldsymbol{\alpha}}) \\
&\quad + \sum_{h_1=1}^{\infty} \sum_{h_2=1}^{\infty} \sqrt{T} \hat{\Delta}_{h_1} \frac{1}{T} \sum_{t=1}^T \mathbf{y}_{t-h_2} \mathbf{y}_{t-h_2}' \sqrt{T} \hat{\Delta}_{h_2}'.
\end{aligned}$$

Since that $T^{-1/2} \sum_{t=1}^T \mathbf{y}_{t-h} \boldsymbol{\varepsilon}_t'(\hat{\boldsymbol{\alpha}}) = O_p(1)$ holds by the martingale CLT theorem under Assumptions 2–3 and $T^{-1} \sum_{t=1}^T \mathbf{y}_{t-h_2} \mathbf{y}_{t-h_2}' = O_p(1)$ holds under Assumption 2, we only need to verify that $\sqrt{T} \hat{\Delta}_h = O_p(1)$ for $1 \leq h \leq p$ and $\sqrt{T} \hat{\Delta}_h = O_p(h\rho^h)$ for $h > p$. Below we will focus on the cases $p \geq p^*, r = r^*, s = s^* = 0$ and $p = p^*, r > r^*, s = s^* = 0$, and the results for other cases can be verified similarly.

When $p \geq p^*, r = r^*$ and $s = s^* = 0$, we have that the model is identifiable from Remark 1 and its proof, and then we can verify $\sqrt{T}(\hat{\boldsymbol{\alpha}}_t - \hat{\boldsymbol{\alpha}}) = O_p(1)$ with similar arguments in the proof of Theorems 1–1. Under Assumption 4, it follows that

$$\begin{aligned}
\sqrt{T} \hat{\Delta}_h &= \sqrt{T}(A_h(\hat{\boldsymbol{\alpha}}_t) - A_h(\hat{\boldsymbol{\alpha}})) = \sqrt{T}(\hat{G}_{t,h} - \mathring{G}_h) = O_p(1) \text{ for } 1 \leq h \leq p, \text{ and} \\
\sqrt{T} \hat{\Delta}_h &= \sqrt{T}(A_h(\hat{\boldsymbol{\alpha}}_t) - A_h(\hat{\boldsymbol{\alpha}})) = \sqrt{T} \sum_{i=1}^r (\hat{\lambda}_{t,i}^{h-p} \hat{G}_{t,p+i} - \mathring{\lambda}_i^{h-p} \mathring{G}_{p+i}) \\
&= \sum_{i=1}^r \left[\hat{\lambda}_{t,i}^{h-p} \sqrt{T}(\hat{G}_{t,p+i} - \mathring{G}_{p+i}) + \sqrt{T}(\hat{\lambda}_{t,i}^{h-p} - \mathring{\lambda}_i^{h-p}) \mathring{G}_{p+i} \right] \\
&= \sum_{i=1}^r \left[\hat{\lambda}_{t,i}^{h-p} \sqrt{T}(\hat{G}_{t,p+i} - \mathring{G}_{p+i}) + \sqrt{T}(\hat{\lambda}_{t,i} - \mathring{\lambda}_i) \sum_{k=0}^{h-p-1} (\hat{\lambda}_{t,i}^{h-p-1-k} \mathring{\lambda}_i^k) \mathring{G}_{p+i} \right] \\
&= O_p(h\rho^h) \text{ for } h > p,
\end{aligned}$$

where $\mathring{\lambda}_i$ and $\mathring{G}_i$ are defined corresponding to $\hat{\boldsymbol{\alpha}}$.

When $p = p^*, r > r^*$ and $s = s^* = 0$, the model is non-identifiable in estimation due to the non-identification of parameters $\{\lambda_i : r^* < i \leq r\}$. However, $\{\lambda_i, G_j : 1 \leq i \leq r^*, 1 \leq j \leq p+r\}$ are identifiable in estimation under Assumption 4, which can be obtained

by Remark 1 and its proof. Thus similar to the proof of Theorems 1, we can show that $\sqrt{T}(\hat{\lambda}_{\iota,i} - \mathring{\lambda}_i) = O_p(1)$ for $1 \leq i \leq r^*$ and $\sqrt{T}(\hat{G}_{\iota,i} - \mathring{G}_i) = O_p(1)$ for $1 \leq i \leq p+r$, where $\hat{\lambda}_{\iota,i}$ (or $\hat{G}_{\iota,i}$) is the LSE or QMLE of λ_i (or G_i) with the order set to ι . Recall that $G_{0i} = 0$ for $p+r^* < i \leq p+r$. Note that $\mathring{\lambda}_i = \lambda_{0i}$ for $1 \leq i \leq r^*$ and $\mathring{G}_i = G_{0i}$ for $1 \leq i \leq p+r$ in this case. Under Assumption 4, it then follows that

$$\begin{aligned}
\sqrt{T}\hat{\Delta}_h &= \sqrt{T}(A_h(\hat{\alpha}_\iota) - A_h(\mathring{\alpha})) = \sqrt{T}(\hat{G}_{\iota,h} - G_{0h}) = O_p(1) \text{ for } 1 \leq h \leq p, \text{ and} \\
\sqrt{T}\hat{\Delta}_h &= \sqrt{T}(A_h(\hat{\alpha}_\iota) - A_h(\mathring{\alpha})) \\
&= \sqrt{T} \sum_{i=1}^{r^*} (\hat{\lambda}_{\iota,i}^{h-p} \hat{G}_{\iota,p+i} - \lambda_{0i}^{h-p} G_{0,p+i}) + \sqrt{T} \sum_{i=r^*+1}^r \hat{\lambda}_{\iota,i}^{h-p} \hat{G}_{\iota,p+i} \\
&= \sum_{i=1}^{r^*} \left[\hat{\lambda}_{\iota,i}^{h-p} \sqrt{T}(\hat{G}_{\iota,p+i} - G_{0,p+i}) + \sqrt{T}(\hat{\lambda}_{\iota,i}^{h-p} - \lambda_{0i}^{h-p}) G_{0,p+i} \right] \\
&\quad + \sqrt{T} \sum_{i=r^*+1}^r \hat{\lambda}_{\iota,i}^{h-p} (\hat{G}_{\iota,p+i} - G_{0,p+i}) \\
&= \sum_{i=1}^{r^*} \left[\hat{\lambda}_{\iota,i}^{h-p} \sqrt{T}(\hat{G}_{\iota,p+i} - G_{0,p+i}) + \sqrt{T}(\hat{\lambda}_{\iota,i} - \lambda_{0i}) \sum_{k=0}^{h-p-1} (\hat{\lambda}_{\iota,i}^{h-p-1-k} \lambda_{0i}^k) G_{0,p+i} \right] \\
&\quad + \sum_{i=r^*+1}^r \hat{\lambda}_{\iota,i}^{h-p} \sqrt{T}(\hat{G}_{\iota,p+i} - G_{0,p+i}) \\
&= O_p(h\rho^h) \text{ for } h > p.
\end{aligned}$$

Thus we have established (A2.32). This together with (A2.31) implies that (a) holds.

Next we show (b). By Lemma A10(i) and Assumption 2, we can obtain that

$$\Sigma(\alpha_0) = E[\varepsilon_t(\alpha_0) \varepsilon_t'(\alpha_0)] + o_p(1) = \Sigma_0 + o_p(1).$$

This together with $\Sigma(\alpha_0) = \Sigma(\mathring{\alpha})$ that we mention above as well as Assumption 3, implies that (b) holds. Thus by (A2.30) and (a)–(b), we have

$$\ln |\hat{\Sigma}(\hat{\alpha}_\iota)| - \ln |\Sigma(\mathring{\alpha})| = \ln(1 + O_p(T^{-1})) = O_p(T^{-1}),$$

that is (A2.29) holds. As a result, (A2.26) holds for Case I.

Case II (misspecification): $p < p^*$ or $r < r^*$ or $s < s^*$. By (A2.27), it holds that

$$\begin{aligned}
\text{BIC}(\iota) - \text{BIC}(\iota^*) &= T \left(\ln |\widehat{\Sigma}(\widehat{\alpha}_\iota)| - \ln |\widehat{\Sigma}(\widehat{\alpha}_{\iota^*})| \right) + (n(\iota) - n(\iota^*)) \ln T \\
&= T \left[\left(\ln |\widehat{\Sigma}(\widehat{\alpha}_\iota)| - \ln |E(\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha}))| \right) \right. \\
&\quad + \left(\ln |E(\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha}))| - \ln |E(\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0))| \right) \\
&\quad + \left. \left(\ln |E(\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0))| - \ln |\widehat{\Sigma}(\widehat{\alpha}_{\iota^*})| \right) \right] \\
&\quad + (n(\iota) - n(\iota^*)) \ln T.
\end{aligned} \tag{A2.33}$$

Hence to establish (A2.26), it suffices to show (c) $\ln |E(\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha}))| - \ln |E(\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0))| > \delta$ for some constant $\delta > 0$; (d) $\ln |\widehat{\Sigma}(\widehat{\alpha}_{\iota^*}) - E(\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0))| = o_p(1)$ as $T \rightarrow \infty$; and (e) $\ln |\widehat{\Sigma}(\widehat{\alpha}_\iota)| - \ln |E(\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha}))| = o_p(1)$ as $T \rightarrow \infty$.

For (c), denote $p_m = \max\{p, p^*\}$, $r_m = \max\{r, r^*\}$, $s_m = \max\{s, s^*\}$ and $\iota_m = (p_m, r_m, s_m)$. And let α_0^+ (or $\dot{\alpha}^+$) be a parameter vector with the order set to ι_m , where α_0 (or $\dot{\alpha}$) being the subvector of α_0^+ (or $\dot{\alpha}^+$) and the extra parameters being zero. Moreover, denote $\mathcal{A}_{\iota_m}$ as the parameter space of α_{ι_m} , including the points α_0^+ and $\dot{\alpha}^+$. Since $E[\varepsilon'_t(\alpha_{\iota^*})\varepsilon_t(\alpha_{\iota^*})]$ has a unique minimum at α_0 on $\mathcal{A}_{\iota^*}$ by Lemma A12, $E[\varepsilon'_t(\alpha_{\iota_m})\varepsilon_t(\alpha_{\iota_m})]$ has a unique minimum at α_0^+ on $\mathcal{A}_{\iota_m}$. This together with $\alpha_0^+, \dot{\alpha}^+ \in \mathcal{A}_{\iota_m}$, $E[\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha})] = E[\varepsilon_t(\dot{\alpha}^+)\varepsilon'_t(\dot{\alpha}^+)]$ and $E[\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0)] = E[\varepsilon_t(\alpha_0^+)\varepsilon'_t(\alpha_0^+)]$, implies that the following result holds for some constant $\delta > 0$:

$$\ln |E[\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha})]| - \ln |E[\varepsilon_t(\alpha_0)\varepsilon'_t(\alpha_0)]| = \ln |E[\varepsilon_t(\dot{\alpha}^+)\varepsilon'_t(\dot{\alpha}^+)]| - \ln |E[\varepsilon_t(\alpha_0^+)\varepsilon'_t(\alpha_0^+)]| > \delta.$$

Thus, (c) holds. Furthermore, (d) holds by Theorem 1(i).

For (e), we only need to show (e1) $\widehat{\Sigma}(\widehat{\alpha}_\iota) - \Sigma(\widehat{\alpha}_\iota) = o_p(1)$; (e2) $\Sigma(\widehat{\alpha}_\iota) - \Sigma(\dot{\alpha}) = o_p(1)$; and (e3) $\Sigma(\dot{\alpha}) - E(\varepsilon_t(\dot{\alpha})\varepsilon'_t(\dot{\alpha})) = o_p(1)$ as $T \rightarrow \infty$. It can be obtained that (e1) holds by Lemma A11, and (e3) holds by the ergodic theorem under Assumption 2. Then we are left to verify (e2). Assume that $E(\varepsilon'_t(\alpha_\iota)\varepsilon_t(\alpha_\iota))$ has a unique minimum at $\dot{\alpha}$ on $\mathcal{A}_\iota$. Similar to the proof of Theorem 1(i), we can show that $\widehat{\alpha}_\iota \rightarrow_p \dot{\alpha}$ as $T \rightarrow \infty$. By the mean value

theorem, we have

$$\|\Sigma(\hat{\alpha}_\iota) - \Sigma(\hat{\alpha})\| \leq \left\| \frac{1}{T} \sum_{t=1}^T \frac{\partial \varepsilon_t(\bar{\alpha}_\iota) \varepsilon'_t(\bar{\alpha}_\iota)}{\partial \alpha} \right\| \|\hat{\alpha}_\iota - \hat{\alpha}\|$$

where $\bar{\alpha}_\iota$ lies between $\hat{\alpha}_\iota$ and $\hat{\alpha}$. This together with $\left\| T^{-1} \sum_{t=1}^T \partial \varepsilon_t(\bar{\alpha}_\iota) \varepsilon'_t(\bar{\alpha}_\iota) / \partial \alpha \right\| = O_p(1)$ by Lemma A10 and $\|\hat{\alpha}_\iota - \hat{\alpha}\| = o_p(1)$ implies (e2) holds. Hence (e) holds. Then by (A2.33) and (c)–(e), we have

$$\text{BIC}(\iota) - \text{BIC}(\iota^*) = T(o_p(1) + \delta) + O(\ln(T)) \rightarrow \infty \text{ as } T \rightarrow \infty.$$

As a result, (A2.26) holds for Case II. This completes the proof of Theorem 2. $\square$

A3 Derivatives and Notations

Recall that $\varepsilon_t(\alpha) = \mathbf{y}_t - \sum_{h=1}^{\infty} A_h(\alpha) \mathbf{y}_{t-h}$ with $\alpha = (\lambda_1, \dots, \lambda_r, \boldsymbol{\eta}'_1, \dots, \boldsymbol{\eta}'_s, \mathbf{g}')'$ and $\boldsymbol{\eta}_j = (\gamma_j, \varphi_j)$. By (2.1), $A_h(\alpha) = G_h$ for $1 \leq h \leq p$, and $A_h(\alpha) = \sum_{k=p+1}^d \ell_{hk}(\boldsymbol{\omega}) G_k$ for $h \geq p+1$ with $\ell_{hk}(\boldsymbol{\omega})$ being the (h, k) -th entry of

$$L(\boldsymbol{\omega}) = \begin{pmatrix} I_p & 0_{p \times (d-p)} \\ 0_{\infty \times p} & \boldsymbol{\ell}^I(\lambda_1) \cdots \boldsymbol{\ell}^I(\lambda_r) \quad \boldsymbol{\ell}^{II}(\boldsymbol{\eta}_1) \cdots \boldsymbol{\ell}^{II}(\boldsymbol{\eta}_s) \end{pmatrix},$$

$\boldsymbol{\ell}^I(\lambda_i) = (\lambda_i, \lambda_i^2, \dots)'$ and

$$\boldsymbol{\ell}^{II}(\boldsymbol{\eta}_j) = \begin{pmatrix} \gamma_j \cos(\varphi_j) & \gamma_j^2 \cos(2\varphi_j) & \cdots \\ \gamma_j \sin(\varphi_j) & \gamma_j^2 \sin(2\varphi_j) & \cdots \end{pmatrix}'.$$

Then the first derivatives of $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ are

$$\begin{aligned}
\frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i} &= -G_{p+i} \sum_{h=p+1}^{\infty} (h-p) \lambda_i^{h-p-1} \mathbf{y}_{t-h}, \\
\frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j} &= \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p-1} \{-\cos[(h-p)\varphi_j] G_{p+r+2j-1} - \sin[(h-p)\varphi_j] G_{p+r+2j}\} \mathbf{y}_{t-h}, \\
\frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j} &= \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p} \{\sin[(h-p)\varphi_j] G_{p+r+2j-1} - \cos[(h-p)\varphi_j] G_{p+r+2j}\} \mathbf{y}_{t-h}, \\
\frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \mathbf{g}'} &= -\mathbf{z}'_t \otimes I_N,
\end{aligned} \tag{A3.1}$$

where $1 \leq i \leq r$, $1 \leq j \leq s$, and $\mathbf{z}_t = \text{vec}(\mathbf{z}_{1t}, \dots, \mathbf{z}_{dt})$ with $\mathbf{z}_{kt} = \sum_{h=1}^{\infty} \ell_{hk} \mathbf{y}_{t-h}$. In addition, $\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) / \partial \mathbf{g}'$ is obtained by rewriting $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ as $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) = \mathbf{y}_t - (\mathbf{z}'_t \otimes I_N) \mathbf{g}$. Furthermore, the second derivatives of $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ are

$$\begin{aligned}
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i^2} &= -G_{p+i} \sum_{h=p+2}^{\infty} (h-p)(h-p-1) \lambda_i^{h-p-2} \mathbf{y}_{t-h}, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j^2} &= \sum_{h=p+2}^{\infty} (h-p)(h-p-1) \gamma_j^{h-p-2} \left\{ -\cos[(h-p)\varphi_j] G_{p+r+2j-1} \right. \\
&\quad \left. - \sin[(h-p)\varphi_j] G_{p+r+2j} \right\} \mathbf{y}_{t-h}, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j^2} &= \sum_{h=p+1}^{\infty} (h-p)^2 \gamma_j^{h-p} \{\cos[(h-p)\varphi_j] G_{p+r+2j-1} + \sin[(h-p)\varphi_j] G_{p+r+2j}\} \mathbf{y}_{t-h}, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \varphi_j} &= \sum_{h=p+1}^{\infty} (h-p)^2 \gamma_j^{h-p-1} \{\sin[(h-p)\varphi_j] G_{p+r+2j-1} - \cos[(h-p)\varphi_j] G_{p+r+2j}\} \mathbf{y}_{t-h}, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \lambda_i \partial \mathbf{g}'_{p+i}} &= - \left[\sum_{h=p+1}^{\infty} (h-p) \lambda_i^{h-p-1} \mathbf{y}_{t-h} \right]' \otimes I_N, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \mathbf{g}'_{p+r+2j-1}} &= - \left\{ \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p-1} \cos[(h-p)\varphi_j] \mathbf{y}_{t-h} \right\}' \otimes I_N, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \gamma_j \partial \mathbf{g}'_{p+r+2j}} &= - \left\{ \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p-1} \sin[(h-p)\varphi_j] \mathbf{y}_{t-h} \right\}' \otimes I_N, \\
\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j \partial \mathbf{g}'_{p+r+2j-1}} &= \left\{ \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p} \sin[(h-p)\varphi_j] \mathbf{y}_{t-h} \right\}' \otimes I_N,
\end{aligned} \tag{A3.2}$$

$$\frac{\partial^2 \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \varphi_j \partial \mathbf{g}'_{p+r+2j}} = - \left\{ \sum_{h=p+1}^{\infty} (h-p) \gamma_j^{h-p} \cos[(h-p)\varphi_j] \mathbf{y}_{t-h} \right\}' \otimes I_N,$$

where $1 \leq i \leq r$, $1 \leq j \leq s$, and $\mathbf{g}_k = \text{vec}(G_k)$ for $1 \leq k \leq d$. The other second derivatives of $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ are zeros.

Recall that $l_t(\boldsymbol{\theta}) = 1/2 \ln |\Sigma(\boldsymbol{\sigma})| + 1/2 \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ and $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \boldsymbol{\sigma}')'$. Note that $\text{vec}(\Sigma) = D \text{vech}(\Sigma)$, where D is an $N^2 \times N(N+1)/2$ duplication matrix. Then the first derivatives of $l_t(\boldsymbol{\theta})$ are

$$\begin{aligned} \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha}} &= \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}} \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}), \\ \frac{\partial l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma}} &= \frac{1}{2} D' \text{vec} \left(\Sigma^{-1}(\boldsymbol{\sigma}) - \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \right), \end{aligned} \quad (\text{A3.3})$$

and the second derivatives of $l_t(\boldsymbol{\theta})$ are

$$\begin{aligned} \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\alpha}'} &= (\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \otimes I_N) \frac{\partial (\text{vec}(\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) / \partial \boldsymbol{\alpha}))}{\partial \boldsymbol{\alpha}'} + \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}} \Sigma^{-1}(\boldsymbol{\sigma}) \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'}, \\ \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\sigma} \partial \boldsymbol{\sigma}'} &= -\frac{1}{2} D' (\Sigma^{-1}(\boldsymbol{\sigma}) \otimes \Sigma^{-1}(\boldsymbol{\sigma}) - \Sigma^{-1}(\boldsymbol{\sigma}) \otimes \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \\ &\quad - \Sigma^{-1}(\boldsymbol{\sigma}) \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha}) \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \otimes \Sigma^{-1}(\boldsymbol{\sigma})) D, \\ \frac{\partial^2 l_t(\boldsymbol{\theta})}{\partial \boldsymbol{\alpha} \partial \boldsymbol{\sigma}'} &= - \left(\boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha}) \Sigma^{-1}(\boldsymbol{\sigma}) \otimes \frac{\partial \boldsymbol{\varepsilon}'_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}} \Sigma^{-1}(\boldsymbol{\sigma}) \right) D, \end{aligned} \quad (\text{A3.4})$$

where the first and second derivatives of $\boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})$ are defined in (A3.1) and (A3.2). In addition, for the $n \times n$ matrices $\mathcal{I} = E[\partial l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta} \partial l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta}']$ and $\mathcal{J} = E(\partial^2 l_t(\boldsymbol{\theta}_0) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}')$. It can be shown that

$$\mathcal{I} = E(P'_{0t} S_1 P_{0t}) \quad \text{and} \quad \mathcal{J} = E(P'_{0t} S_2 P_{0t}), \quad (\text{A3.5})$$

where the $(N + N^2) \times n$ matrix $P_{0t} = P_t(\boldsymbol{\theta}_0)$, and the $(N + N^2) \times (N + N^2)$ matrices

$$S_1 = \begin{pmatrix} \Sigma_0^{-1} & -\frac{1}{2} \Sigma_0^{-1} E(\boldsymbol{\varepsilon}_{0t} \mathbf{u}'_{0t}) \mathcal{P} \\ * & \frac{1}{4} [\mathcal{P} E(\mathbf{u}_{0t} \mathbf{u}'_{0t}) \mathcal{P} - \text{vec}(\Sigma_0^{-1}) \text{vec}(\Sigma_0^{-1})'] \end{pmatrix} \quad \text{and} \quad S_2 = \begin{pmatrix} \Sigma_0^{-1} & 0_{N \times N^2} \\ * & \frac{1}{2} \mathcal{P} \end{pmatrix},$$

with $*$ denoting the symmetric elements, $\mathcal{P} = \Sigma_0^{-1} \otimes \Sigma_0^{-1}$, $\mathbf{u}_0 = \text{vec}(\boldsymbol{\varepsilon}_{0t}\boldsymbol{\varepsilon}_{0t}')$ and

$$P_t(\boldsymbol{\theta}) = \begin{pmatrix} \frac{\partial \boldsymbol{\varepsilon}_t(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}'} & 0_{N \times N(N+1)/2} \\ 0_{N^2 \times n_\alpha} & D \end{pmatrix}.$$

A4 The Connection between the Scalable ARMA and VARMA Models

We first consider the VARMA(1,1) model

$$\mathbf{y}_t = \Phi \mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t - \Theta \boldsymbol{\varepsilon}_{t-1}, \quad (\text{A4.1})$$

where $\Phi, \Theta \in \mathbb{R}^{N \times N}$. Suppose that all eigenvalues of Θ are less than one in absolute value. (A4.1) can be written as the following VAR(∞) process:

$$\mathbf{y}_t = \sum_{h=1}^{\infty} A_h(\Phi, \Theta) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t \quad \text{with} \quad A_h(\Phi, \Theta) = \Theta^{h-1}(\Phi - \Theta). \quad (\text{A4.2})$$

Then we reparameterize A_h 's in (A4.2). Suppose that Θ has r nonzero real eigenvalues, $\lambda_1, \dots, \lambda_r$, and s conjugate pairs of nonzero complex eigenvalues, $(\lambda_{r+2j-1}, \lambda_{r+2j}) = (\gamma_j e^{i\varphi_j}, \gamma_j e^{-i\varphi_j})$ for $1 \leq j \leq s$ with i being the imaginary unit. Θ can be block-diagonalized as $\Theta = BJB^{-1}$ via the Jordan decomposition, where B is an invertible matrix and $J = \text{diag}\{\lambda_1, \dots, \lambda_r, C_1, \dots, C_s, 0'_{(N-r-2s) \times 1}\}$ with each C_j being the 2×2 block determined by $\boldsymbol{\eta}_j = (\gamma_j, \varphi_j)$ as follows

$$C_j = \gamma_j \begin{pmatrix} \cos(\varphi_j) & \sin(\varphi_j) \\ -\sin(\varphi_j) & \cos(\varphi_j) \end{pmatrix}.$$

Substituting $\Theta = BJB^{-1}$ into (A4.2), we obtain that

$$\mathbf{y}_t = \sum_{h=1}^{\infty} B J^{h-1} B^{-1} (\Phi - \Theta) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t.$$

Let $\underline{A} = B^{-1}(\Phi - \Theta)$ and $\underline{B} = B$. It holds that

$$\mathbf{y}_t = (\Phi - \Theta)\mathbf{y}_{t-1} + \sum_{h=2}^{\infty} \underline{B}J^{h-1}\underline{A}\mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t.$$

Furthermore, according to the block form of J , let $\underline{A} = (\underline{A}_1, \dots, \underline{A}_r, \underline{A}_{r+1}, \dots, \underline{A}_{r+s}, \underline{A}_{r+s+1})'$ and $\underline{B} = (\underline{B}_1, \dots, \underline{B}_r, \underline{B}_{r+1}, \dots, \underline{B}_{r+s}, \underline{B}_{r+s+1})$ with $\underline{A}_i = \mathbf{a}_i$ and $\underline{B}_i = \mathbf{b}_i$ being $N \times 1$ matrices for $1 \leq i \leq r$, $\underline{A}_{r+j} = (\mathbf{a}_{r+j}, \bar{\mathbf{a}}_{r+j})$ and $\underline{B}_{r+j} = (\mathbf{b}_{r+j}, \bar{\mathbf{b}}_{r+j})$ being $N \times 2$ matrices for $1 \leq j \leq s$, and $\underline{A}_{r+s+1}$ and $\underline{B}_{r+s+1}$ being $N \times (N - r - 2s)$ matrices. It then follows that

$$\begin{aligned} \mathbf{y}_t &= (\Phi - \Theta)\mathbf{y}_{t-1} + \sum_{h=2}^{\infty} \left[\sum_{i=1}^r \lambda_i^{h-1} \mathbf{b}_i \mathbf{a}'_i + \sum_{j=1}^s (\mathbf{b}_{r+j}, \bar{\mathbf{b}}_{r+j}) C_j^{h-1} (\mathbf{a}_{r+j}, \bar{\mathbf{a}}_{r+j})' \right] \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t \\ &= (\Phi - \Theta)\mathbf{y}_{t-1} + \sum_{h=2}^{\infty} \left\{ \sum_{i=1}^r \lambda_i^{h-1} \mathbf{b}_i \mathbf{a}'_i + \sum_{j=1}^s \gamma_j^{h-1} \left[\cos((h-1)\varphi_j) (\mathbf{b}_{r+j} \mathbf{a}'_{r+j} + \bar{\mathbf{b}}_{r+j} \bar{\mathbf{a}}'_{r+j}) \right. \right. \\ &\quad \left. \left. + \sin((h-1)\varphi_j) (\mathbf{b}_{r+j} \bar{\mathbf{a}}'_{r+j} - \bar{\mathbf{b}}_{r+j} \mathbf{a}'_{r+j}) \right] \right\} \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t. \end{aligned} \tag{A4.3}$$

Let $G_1 = \Phi - \Theta$, $G_{1+i} = \mathbf{b}_i \mathbf{a}'_i$ for $1 \leq i \leq r$, and $G_{1+r+2j-1} = \mathbf{b}_{r+j} \mathbf{a}'_{r+j} + \bar{\mathbf{b}}_{r+j} \bar{\mathbf{a}}'_{r+j}$ and $G_{1+r+2j} = \mathbf{b}_{r+j} \bar{\mathbf{a}}'_{r+j} - \bar{\mathbf{b}}_{r+j} \mathbf{a}'_{r+j}$ for $1 \leq j \leq s$. We can rewrite (A4.3) in the form of scalable ARMA model in (2.1), that is,

$$\mathbf{y}_t = \sum_{h=1}^{\infty} A_h(\boldsymbol{\omega}, \mathbf{g}) \mathbf{y}_{t-h} + \boldsymbol{\varepsilon}_t,$$

where $A_1(\boldsymbol{\omega}, \mathbf{g}) = G_1$ and $A_h(\boldsymbol{\omega}, \mathbf{g}) = \sum_{i=1}^r \lambda_i^{h-1} G_{1+i} + \sum_{j=1}^s \gamma_j^{h-1} [\cos((h-1)\varphi_j) G_{1+r+2j-1} + \sin((h-1)\varphi_j) G_{1+r+2j}]$ for $h \geq 2$.

For the general VARMA(p, q) model

$$\mathbf{y}_t = \sum_{i=1}^p \Phi_i \mathbf{y}_{t-i} + \boldsymbol{\varepsilon}_t - \sum_{j=1}^q \Theta_j \boldsymbol{\varepsilon}_{t-j}, \tag{A4.4}$$

where $\Phi_i, \Theta_j \in \mathbb{R}^{N \times N}$ for $1 \leq i \leq p$ and $1 \leq j \leq q$. Suppose that (A4.4) is invertible, then

it can be written as the following VAR(∞) process:

$$\mathbf{y}_t = \sum_{h=1}^{\infty} \underbrace{\left(\sum_{i=0}^{p \wedge h} P \underline{\Theta}^{h-i} P' \Phi_i \right)}_{A_h} \mathbf{y}_{t-h} + \varepsilon_t$$

with

$$\underline{\Theta} = \begin{pmatrix} \Theta_1 & \Theta_2 & \cdots & \Theta_{q-1} & \Theta_q \\ I_N & 0_{N \times N} & \cdots & 0_{N \times N} & 0_{N \times N} \\ 0_{N \times N} & I_N & \cdots & 0_{N \times N} & 0_{N \times N} \\ \vdots & \vdots & & \vdots & \vdots \\ 0_{N \times N} & 0_{N \times N} & \cdots & I_N & 0_{N \times N} \end{pmatrix},$$

where $\Phi_0 = -I_N$ and $P = (I_N, 0_{N \times N(q-1)})$ are constant matrices. Using the same techniques, we can also rewrite (A4.4) in the form of scalable ARMA model in (2.1); see also the proofs of Proposition 2.2 of [Huang et al. \(2025\)](#) and Proposition 1 of [Zheng \(2025\)](#).

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